IR CHECKLIST

1. BASIC INFORMATION	
Process	
Project name	
Customer	
Application	
Product manager	
IP designer	
QA manager	
Integration review date	
MTO Schedule	

2. DESIGN CHECK SHEET

Recommendation : when you fill the integratio

CATEGORY	ITEM
Fundamental information	Temperature & Reliability check condition
IP Version check	IP DK Version
Order form re-check	Order Form
	HTOL
	Power up check
	Start up check
	Power down check

Function check	Function simulation
	Function simulation
	Function simulation
	Function simulation
	SFR Setting Check
	Function - ATE(Automatic Test Equipment) Vector Test
	Function simulation
	Function simulation
	Unused pin check
	On-chip LDO Setting
	APB Interface

	APB _ Write Transfer
	APB _ Read Transfer
	APB Address for internal Registers
	IDDQ Mode
	Level Shifter
	Level Shifter

Implementatiaon check	**Timing Skew Check for Direct Access Port in Test Mode**
	STA(Static Timing Analysis)
	Transmitter input Timing
	Network Jitter
	Network Jitter
	Network Jitter
	Product floor plan
	PAD

Floor plan	**PAD**
	PAD
	IP Placement
	Bumpmap
	I/O ring check
	Breakcell check
	Linker cell check
	Back to back diode (Anti-parallel diode)

Integration check	Important signal shielding
	Verification rule check
	IO Connection Check
	Top layout review
	ESD
	RDL
	PDN
	PDN
	PDN

	Power & Ground routing
	Power routing
	ESD
	ATB (Analog Test Bus)
	Shielding
	Inptu & Output Clock Connection
Verification check	**Device List check**
Off-Chip	**Package Ground Plane Merge**
	Package Sign-Off

On-chip	**Ballmap**

	COLOR DESC
	I
	For Project Manag
	For Logical
	For Physical Designer(
	For Physical Designer(

n review checklist, please follow the guide in IP datasheet and userguide

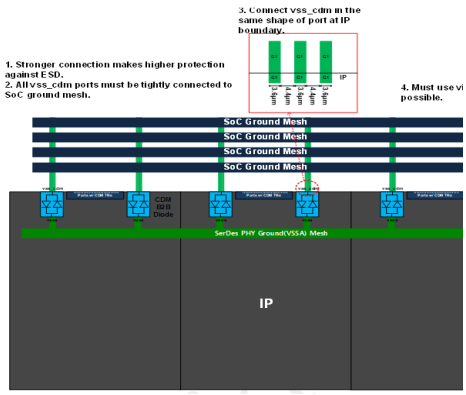
CHECK POINT	CHECK
Is the chip operating temperature condition and life cycle checked?	Review and make comments - Chip operating environment (mobile, au - Junction temperature condition - The reliability(EM) cycle(time)
Check whether the version of the DB used is final (FE/BE/DOC)	Have you used the latest version of DK frc Enter the name of the used DK in column
Are specification consensus done with order form between customer and IP designer?	Please attach the final OrderForm doc. wh
In full-chip level HTOL(SCAN) test mode, have you checked that the IP's inputs are set as conditions guaranteed by the UserGuide? Have you verified that the IP operates without any problem by simulation	1) Please share dump file path and file na 2) Please share the IP's hierarchical inform 3) Please share used waveform view & ve
Is power-management sequence which is supplied to IP guaranteed in SoC?	Please share the figure of the power sequ - If you share a dump file, please share file information. - If there is no figure or dump file, please ALWAYS ON, ALIVE, etc.).
IP start-up sequence check	Have you checked that the clock input an start-up sequence described in the Datash
Power down sequence check	Have you checked that the power sequen sequence described in the Datasheet?

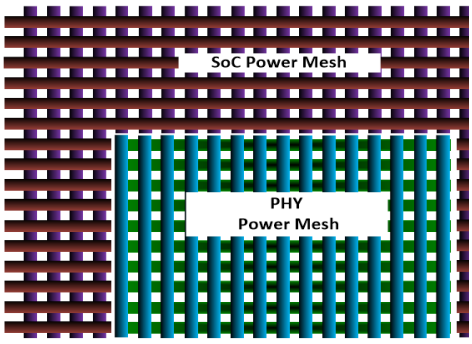
Input control sequence during the operation	Have you checked If there is the case that enabled and in normal operation? Is there operation scenario with the verilog model
Are the verification results in product level shared with IP designer? (IP normal scenario; RTL simulation or post simulation results under the actual product scenario) Please check whether appropriate control and default value are applied through those Verilog simulations.	1) Please share dump file path and file na 2) Please share the IP's hierarchical inform 3) Please share used waveform view & ve
Stand-by scenario during lock-time	If the IP is a clock IP which outputs the tar checked whether this is reflected to the sy waits for the lock time/settling time as spe
Test mode verification	Have you checked that, in the test mode f maximum frequency output to I/O is withi I/O? Write the maximum frequency output in c
Are SFR setting values applied as the guide which is provided by IP designer?	Please reviews it and attach confirmation
Are function test results reviewed? (All about IP's test mode) Please check whether appropriate control and default value are applied through those Verilog simulations.	1) Please share dump file path and file na 2) Please share the IP's hierarchical inform 3) Please share used waveform view & ve
Input controllability	Have you checked if there is any case whe ROM code or tie cell and cannot be easily If there is, write the list of IP's inputs conn 'Note'
Input controllability	Have you verified the cases when IP is cor and checked there is no problem?
Are there any IP input/output pins that are not used? All IP inputs except the unused-pin list must be controllable.	1) Please specify about the pin-names tha 2) Please write the logical states of the un gnd with logic block, floating) 3) If all pins are controllable, please write
LDO setting	In case IP power is received from another information so that stable voltage is suppl -. If LDO trim is needed, has the appropria the LDO? -. If the LDO uses the chopping mode, is tl according to the guide?
Check for interface frequency	The APB interface is designed for low band check or select the maximum interface fre

Check for write sequence	Has the write transfer been set up correct '-. i_apb_paddr, i_apb_pwrite, i_apb_psel a after the rising edge of the i_apb_pclk. -. After the following edge of the i_apb_pc -. i_apb_paddr, i_apb_pwdata and i_apb_p throughout the access cycle and the tra cycle. -. i_apb_penable signal is deasserted at th signal proceeds to low. -. To reduce power consumption, i_apb_pa alter.
Check for read sequence	Has the read transfer been set up correctl -. i_apb_paddr, i_apb_pwrite and i_apb_ps edge of the i_apb_pclk. -. After the following edge of the i_apb_pc -. The slave provides the o_apb_prdata du -. The data is sampled on the rising edge cycle.
Check for APB Upper Byte to SerDes Macro mapping	The bit[31:14] of APB address must be tie
Check for IDDQ mode operation	Has the IDDQ mode operation been refer '-. Drive i_scan_rstn and i_ropll_scan_rstn -. Drive all other PHY interface input signa -. Drive i_scan_mode[1:0] and i_ropll_scan
If level-shifter(LS) is used outside IP, share where it used and how connected with PWR and GND. The LS must be inserted if the supply source is different, even if the supply voltage level is same between the IP and SOC blocks.	Read "1.3.5 Level shifter guide" from the Level shifters must be inserted to all conn reference clock ports such as refclk_xtal a I/O cells through level shifters. The PHY-si connected to vdd (vdda_d from Table 3-1) 1) Please write cell name and input & outp 2) Specify pin names of IP connected to e 3) Please share vslp(vclp) report or figure of external level-shifter.
Impact analysis of using level shifter (if used)	If there is any external level shifters used jitter/duty degradation with simulation? If level shifter is used at the clock output, in column J, 'Note'

Are timing skew for ports which access directly from GPIO checked for direct access (stand-alone) test? When in PHY Test mode, the APB interface is accessed immediately from the GPIO. At this time, check whether skew is considered for Multi bit. (For example, PADDR (16bit), RDATA, PWDATA (16bit), PSEL, PENALBE, PWRITE) If there is a difference in skew, eventually the ATE must turn on the timing to skew. In fact, it is difficult to consider skew in the ATE. In the Post STA result, the skew report is received separately.	Please attach the report of the time delay checking timing skew between ports for A Use skew report from post STA, about IN/(Please, refer IP user guide to check target
Are STA results between IP and product reviewed? 1) Review the unconstraint path with the IP designer (The important timing path must be constraint path) 2) Check the timing report (setup / hold timing reports)	1) Please share STA Report: Refer to Data interest with respect to STA (e.g. FIN, FOU 2) Plase share the unconstraint path list
Check input timing	Read "1.5.1 Transmitter Input Timing" fro **The TX input signals are timed from r** recommended that the ASIC(or SOC) uses clock the parallel TX data shipped to PMA. derived from o_tx_link_sym_clk clock. And int_o_tx_link_sym_clk which is internal clo
Are the clock network jitters from full chip shared and reviewed with IP designers?	Please attach the result of the network jitt SOC to the IP's input clock pin (e.g. ref_clk
Low jitter path management	When the output of PLL is injected to the i ADC, which need low jitter performance, h PLL output and the receiving IP is minimiz as possible?
RJ/DJ simulation	You should execute RJ/DJ simulation for lo Write RJ and DJ results in column L."Note".
Is the floorplan shared with IP designer in advance?	Please share product floorplan that shows
I/O pad type	Have you checked that appropriate type c I/O in column J, 'Note'

Isolation of I/O pad	Is the Power I/O for the IP properly separa cells? Have you reviewed over any ESD is ESD reviewer in column J, 'Note'.
Noise around I/O pad	Have you checked that there is no noisy s around the IP's Power I/O?
Is minimum distance between IP and the other glue logic to ensure no DRC violation secured? What is the distance between IP and noisy or high power consuming IPs (e.g. DDRPHY)?	Is the IP placed properly based on "1.3 La -. rs_hsstptx2ch_ln14lpp should be placed minimize package R/L/C parasitic -. In case rs_hsstptx2ch_ln14lpp cannot be enough space (e.g. > 1000um) from the -. rs_hsstptx2ch_ln14lpp is also recommer switching logic. The distance to other IP as 20µm and 40µm, respectively. -. If there are noisy blocks such as DDR PH side of the die (e.g. the opposite side). T the DDR PHY may cause unwanted nois package/board level. To prevent unwant noisy block at the opposite side in the d -. For the process monitoring block placen located near the PHY. We recommend y block within 100um from the PHY and cl (e.g. < 400 um from chip edge). This bl minimum space rule of 5um from our PH um space) because this block is not wor Please specify the distance or attach a ca
Is bumpmap of each IP shared and reviewed by IP designer?	Please share bumpmap file or figure. If there is no Bump, please write N/A.
Is RTO/SNS/VSS ring connected without breaking in IP's LEF and GDS? Are the defined as ports in IP's LEF and GDS?	Please check and specify the RTO/SNS/VS GPIOs
Are the breakcells on I/O ring on top level connected properly according to guide?	If breakcells are used, please specify the
Is linker cell used beside I/O ring of IP?	If linker cells are used, please specify the
Are all the ground pins connected to the logic ground in IP connected by metal according to pin width? (e.g. VSS_CDM port which is back-to-back diode output on IP connected to the ground of the product by proper metal and width)	Please share the layout captured image o 1) SOC logic ground should be connected on IP. 2) Must be connected with bar-via. These are very important items to guaran latch-up.

Are important input/output signals of IP shielded in full chip level?	signals. And please share the shielding informatio 1) Clock shielding ratio: (shielded line lenc
Are there any DRC, EAC or PERC errors of IP in chip level? If there are, are these checked and waived by IP designer?	Please review it and attach the waive-list. If there is no error regarding IP, please wr
Check the minimum routing resistance and skew, if the IP does not include IO. (ex: LVDS, USB, DCPHY)	1) Please attach the layout capture for rou 2) Please attach the layout capture for dif
Is the final version of top layout of the full chip shared? Is it reviewed by the IP designer whether there is no issue in physical integration on IP level inputs, outputs, IOs, power supplies, ground to full chip level connections, IP space?	Please choose one among two methods in - Please share the PPT about figures of IP' - Please share the layout DB path(OPUS p top. This path must be accessible by the I
Back-to-back diode connection	In order to prevent CDM damage between supplies, back-to-back diodes are inserted rs_hsstptx2ch_ln14lpp should be strongly Figure 1-6. Thicker metal layers like Kx or r 
RDL routing crosstalk	Have you made sure that no "signal RDL" thumb is that nothing is allowed to go ove itself or the power RDL of the IP itself.
PDN analysis - execution/review	Have you reviewed the power noise of the
Check the level of voltage ripple	Have you checked that the voltage ripple result of PDN analysis are satisfied? -. High power supply voltage for the PHY (-. Low power supply volatage for the PHY
Check whether PDN analysis results are considered in integration	Have you added sufficient on-chip De-cap PDN analysis result? Write the size[F] of inserted de-cap in colu specification from PDN analysis could be r please write 0[F].

Prevention power supply noise	Has the power & ground routing been des 1.3.3 power and ground routing? -. In order to prevent power supply noise v lines, rs_hsstptx2ch_ln14lpp's power an those of the SoC as illustrated in followi 
Power routing connection - width/metal	Have you connected the power/ground ro defined at the IP port? Or did you connect (which is also an allowed case)?
ESD check	Have you connected the CDM/Clamp/IO ac Userguide "1.3.4 ESD Protection", and got final version? Or is there any tool (PERC/DRC/Pathfinder the ESD reviewer in column J, 'Note'.
ATB Connection	Read "1.3.6 ATB Connection" from the Use The port atb can be treated in three ways. • No connection when it is not used • Connection to I/O cell and bump but r • Connection to I/O cell, bump and PKG The port atb must be floated in the case o if the ATB port is not connected, has it bee
Shielding of major signals	Read 1.3.7 Shielding Guide" from the Use Have you checked that "two reference clo shielded by adjacent metals"
Check input & outpunt clock declaration	Read "1.5.2 Input and Output clcok declar Have you properly routed the input/outpu
Check if all devices in IP are available in the product.	Please check device list with this method - In IP BE-Common, LVS report(lvsprt_ca.lv Product designer should check if all of the is an issue, please contact IP designer. - Product designer share the device list of
Is the ground of the IP merged with ground plane of SOC in package level?	Please review the ground ball name in bal attach the confirm material
Is the package sign-off confirmed by DT(Design technology) and package team?	Please review and attach the results

Are ballmaps for IP reviewed with IP designer? If there is an fixed signal order of connector, are signal balls correctly placed?	Please attach ball-maps and review confir If that IP use connector, please fill in signa

CRIPTION (ITEM)		
tem	PASS	
er or SOC (Block Top)	NG	
Designer(RTL)	N.A	
(Layout - Physical Design)	Not yet	
(Physical Implementation)	SCORE	
	100%	
	80~99%	IR ca

.

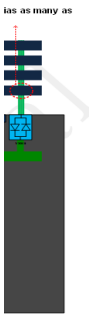
K METHOD	PRO	
	CHECK RESULT	NAME/DATE (YY.MM.DD)
tomotive, network)	Not yet	
om Rasmschip? J, 'Note'	Not yet	
ich are confirmed.	Not yet	
me nation rsion	Not yet	
ence or dump file. e path / file name / IP's hierarchical make comments about power sequence(e.g.	Not yet	
d control signals are applied according to the heet?	Not yet	
ce applied in the product follows the power	Not yet	

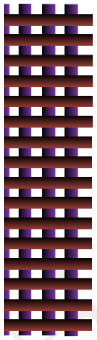
t input control signals change while the IP is any problem? Have you simulated the ?	**Not yet**	
me nation rsion	**Not yet**	
get frequency after a certain time, have you ystem implementation where the system ecified in the Datasheet?	**Not yet**	
for evaluating the IP's characteristics, the n the specified maximum operating speed of olumn J 'Note'	**Not yet**	
email from IP designer	**Not yet**	
me nation on fullchip rsion	**Not yet**	
ere specific inputs of IP are connected to a modified by SW modification? nected to ROM code or signal in column J,	**Not yet**	
nnected to ROM/Tie cell by Verilog simulation	**Not yet**	
t are not used or not controllable. used pins. (e.g. tie to vdd with tie-cell, tie to "All pins are used and controllable" with the	**Not yet**	
LDO IP, have you checked the following lied during IP operation? ate trim code value been applied to he chopping clock supplied	**Not yet**	
dwidth and low speed. Ensure that you do not equency	**Not yet**	

:ly? ınd i_apb_pwdata signals are modified lk, i_apb_penable signal is asserted. sel signals are all remain valid nsfer completes at the end of the ıe end of the transfer. The i_apb_psel ıddr and i_apb_pwrite signal does not	**Not yet**	
ly? el signals are modified after the rising lk, i_apb_penable signal is asserted. ring the access cycle. of i_apb_pclk at the end of the access	**Not yet**	
d to 0.	**Not yet**	
to the below? to 0 ls to 0 ı_mode to 1	**Not yet**	
Userguide. ıection between the PHY and SoC. Two ınd refclk_soc_pll also must be connected to de supply port of level shifters must be ı of the PHY. ɔut voltage domain of external level shifter. xternal level shifter. or GDS to check power & ground connections	**Not yet**	
at the clock output, have you checked write the amount of jitter/ duty degradation	**Not yet**	

/ from GPIO to target ports of IP (This is for .TE) OUT multi-bit. : pins and timing requirements.	**Not yet**	
sheet & Userguide to check IP's pin name of JT)	**Not yet**	
m the Userguide. **rising edge of i_tx_data_clk.** It is : the o_tx_link_sym_clk (or divide-by 2) to . In other words, the i_tx_data_clk should be l o_tx_link_sym_clk is generated clock of ck pin.	**Not yet**	
:er simulation(PJX) from the clock source on k_in, FIN, etc.).	**Not yet**	
input of other mixed signal IPs like PHY or have you checked that clock path between the ed and path distance is routed as the shortest	**Not yet**	
w jitter clock path.	**Not yet**	
: the location of all IPs	**Not yet**	
of I/O are used? Write the name of the used	**Not yet**	

ted from noise interference around using slot sues with ESD team? Write the name of the	**Not yet**	
ources (i.e. Tx Driver/Signal I/O) placed	**Not yet**	
yout guide from userguide? in the center of one-side to e placed in the center, there should be e chip edge. nded to be far away from noisy and logic blocks is recommended HY, then place this IP at the other The noisy switching IO signals from e coupling in the die and/or ted notice, ensure that you place the ie. nent, at least one block should be ou to place the process monitoring loser to the IO region of the PHY ock can be located with the HY (e.g. 50 ~ 300 um space → 5 king in the normal mode. ptured image	**Not yet**	
	Not yet	
S's ring and ports in case of IP including	**Not yet**	
cell name	**Not yet**	
cell name	**Not yet**	
f the connected by each metal layer. with ground of IP as metal width of pin(port) tee ESD specifications like HBM, CDM, and	**Not yet**	

n of signal lines. ıth/total line length)*100	**Not yet**	
ite comment "There is no error regarding IP".	**Not yet**	
ıting between the core and I/Os. ferential signal routings.	**Not yet**	
ı below. s integration with details. ath) that allows access to layout view of the P designer.	**Not yet**	
ı rs_hsstptx2ch_ln14lpp and SoC power l in the PHY. The "vss_cdm" ports on connected to SoC grounds as illustrated in Gx with enough number of vias are highly  	**Not yet**	
of other block(s) pass over the IP? Rule of ɜr the IP except the power bump of the IP	**Not yet**	
ɜ IP and the reviewed the PDN results?	**Not yet**	
specification for each IP allowed as a the Core : under 30mVpp Core: under 20mVpp	**Not yet**	
between IP's power and ground based on ımn M, 'Note'. If the voltage ripple met without additional de-cap insertion,	**Not yet**	

igned according to the userguide which induces jitter at the TX serial d ground should be isolated from ng 	**Not yet**	
utings by using the same metal layer/width as them with wider metal/more metal stack	**Not yet**	
ccording to the guidelines described in the confirmed by ESD team after reviewing the etc.) verification result? Write the name of	**Not yet**	
erguide. . no PKG ball assign ball of non-connection. en as floating?	**Not yet**	
rguide. ck pins refclk_xtal and refclk_soc_pll must be	**Not yet**	
ation guide" from the Userguide. t clock signals?	**Not yet**	
in below. vs) file includes IP device list. devices are available in the product. If there product, then IP designer would check it.	**Not yet**	
l map and where the merging point is and	**Not yet**	
	Not yet	

mation emails from IP designer. al order of connector	**Not yet**	

SCORE		
0	PASS	
0	NG	
0	N.A	
55	Not yet	
0%	SCORE	
PASS	50~79%	I
n be determined as marginal pass	Under 50%	IR

DUCT (PROJECT)	IP	
NOTE	CHECK RESULT	NAME/DATE (YY.MM.DD)
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ESD 검토.eml	**Not yet**	
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ram-sCHIP
FNALOGIXEDIGNAL

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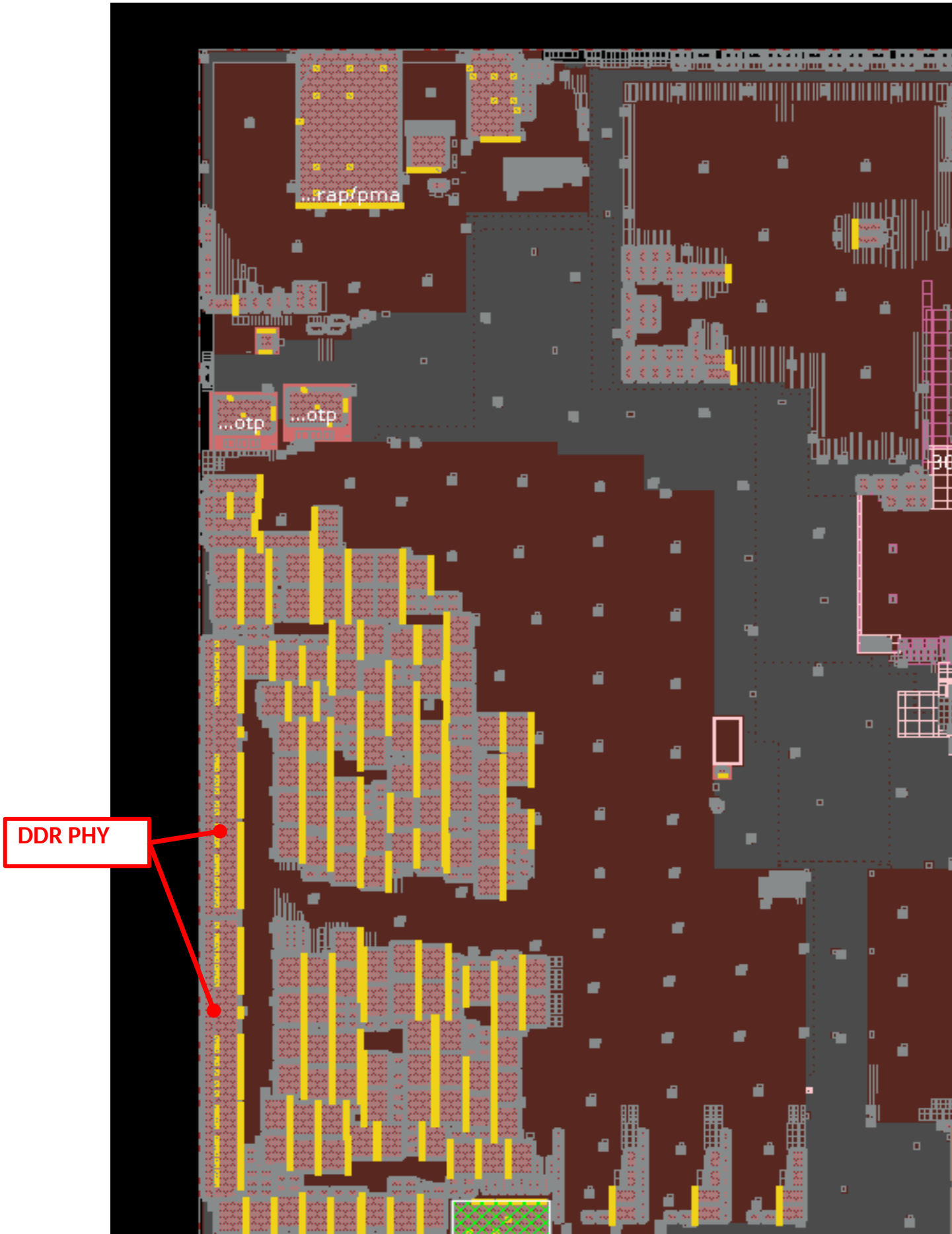
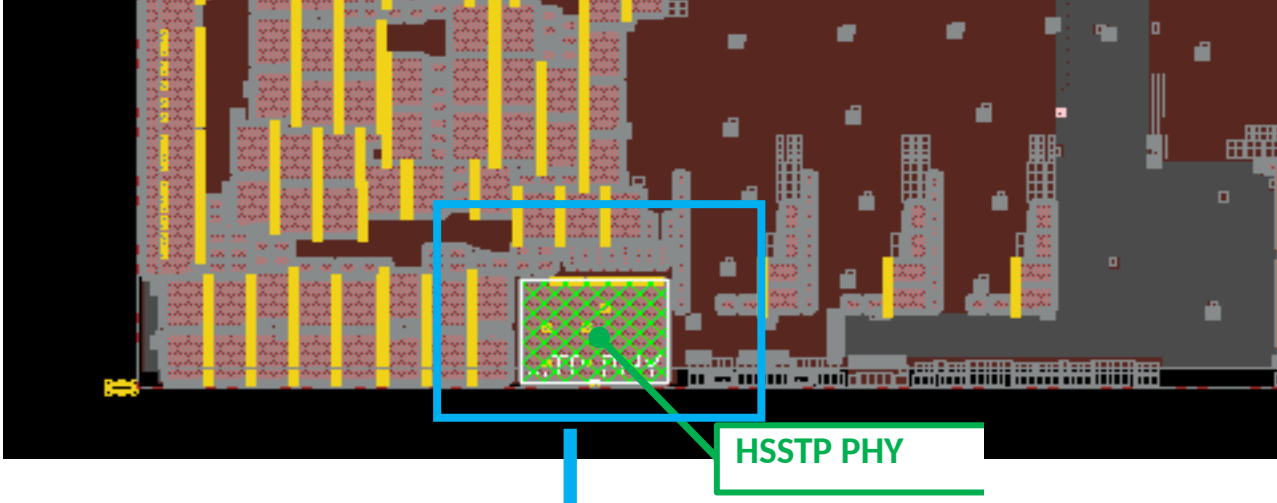
R can be started
can not be started

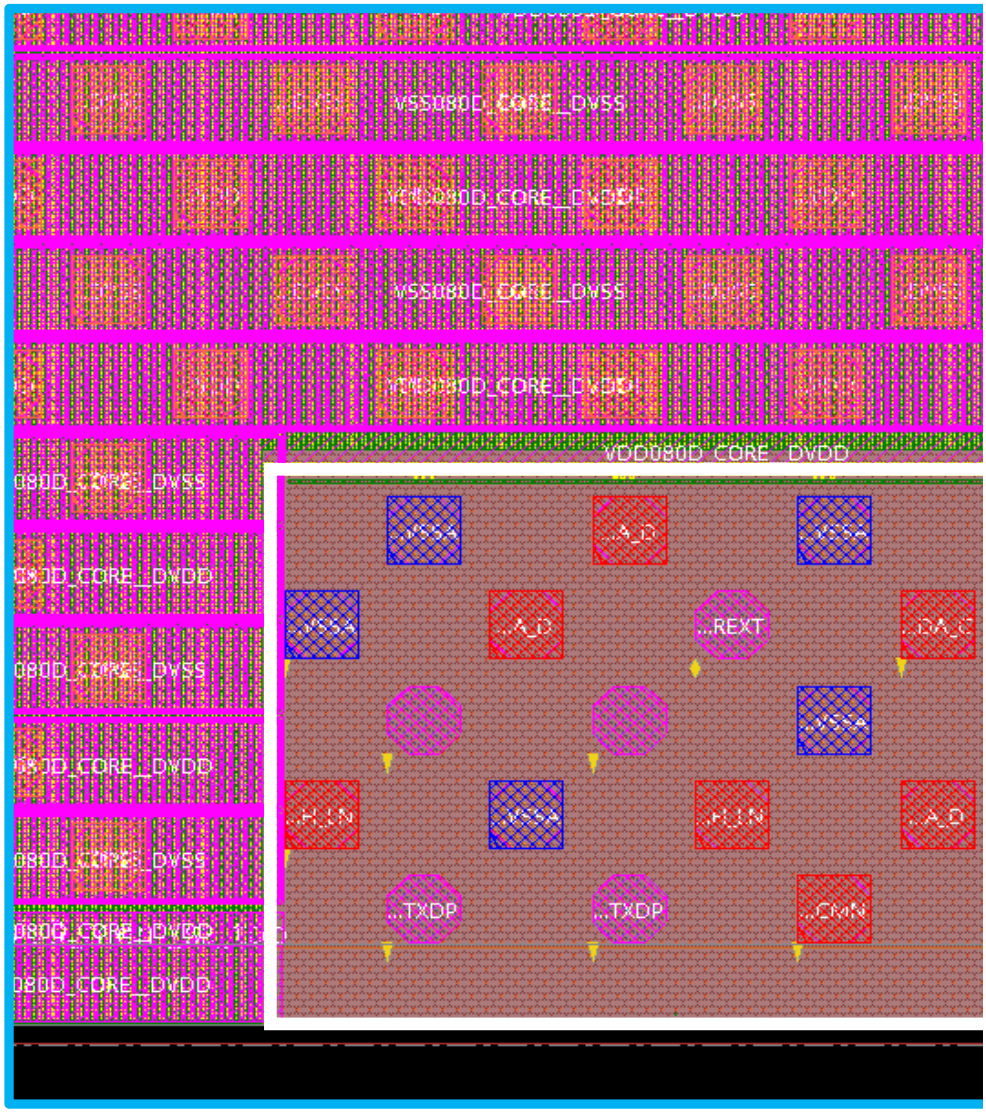
NOTE

Product floor plan

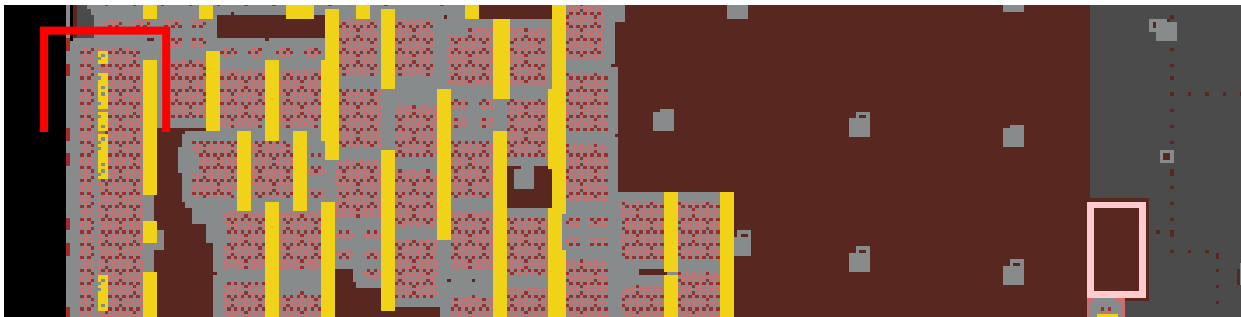


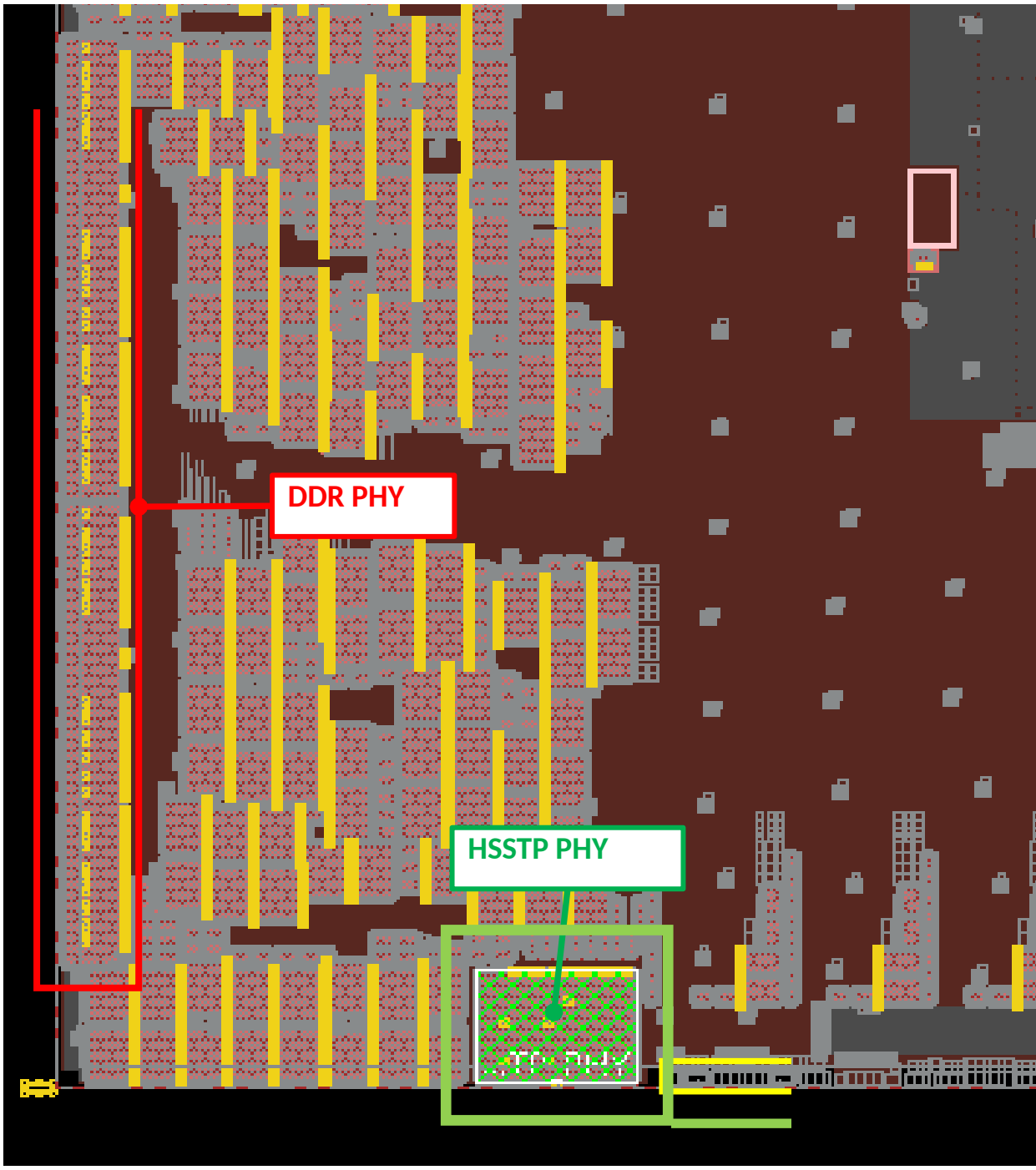


PAD

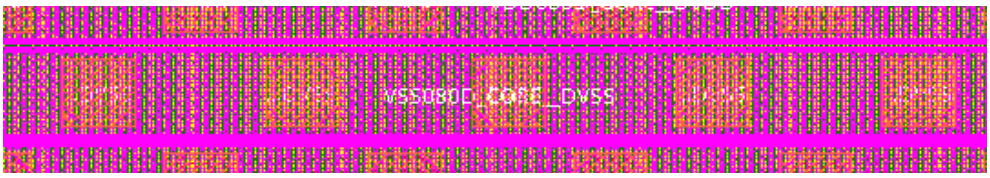


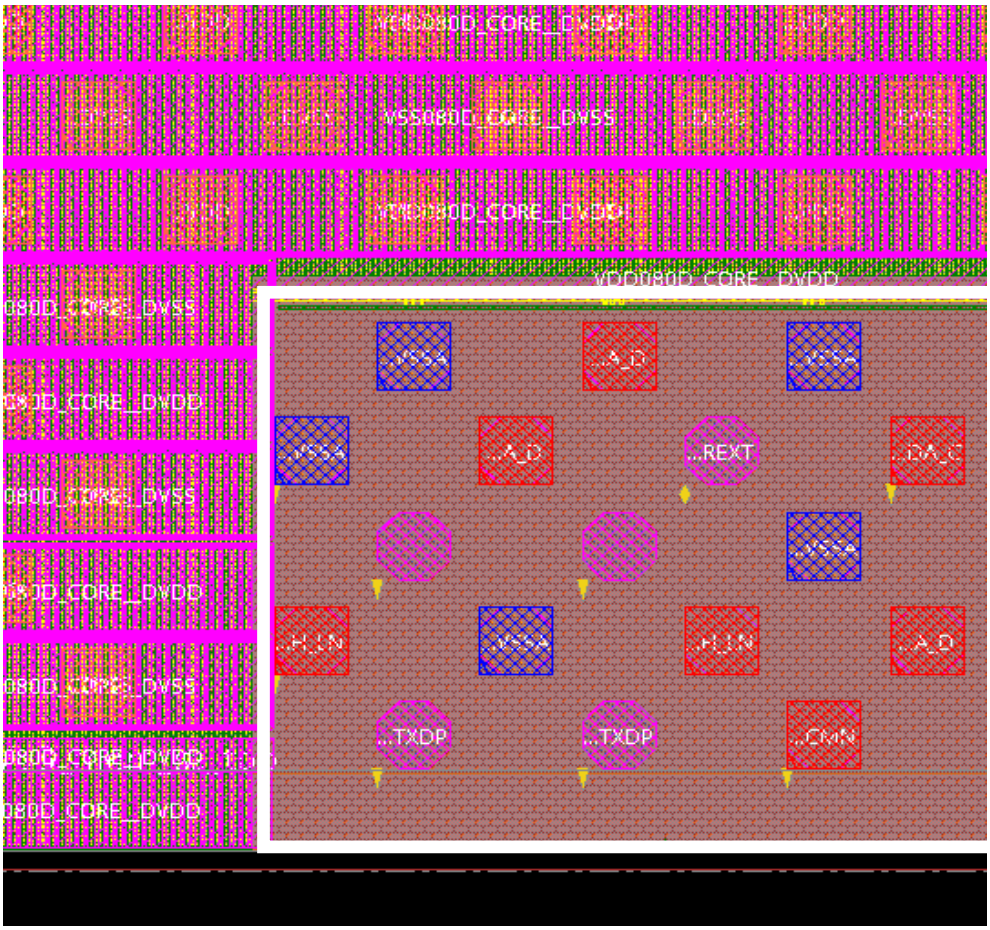
IP Placement



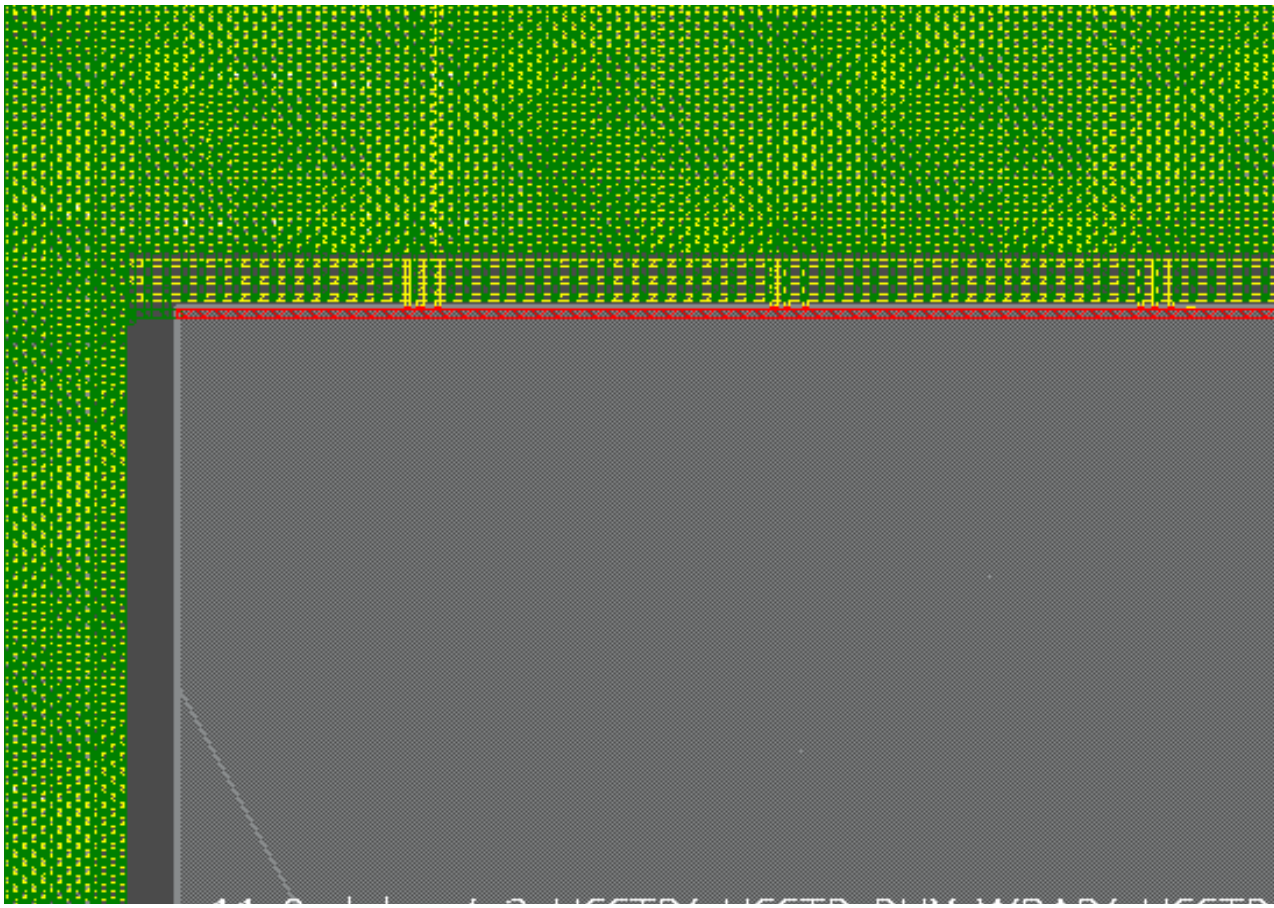


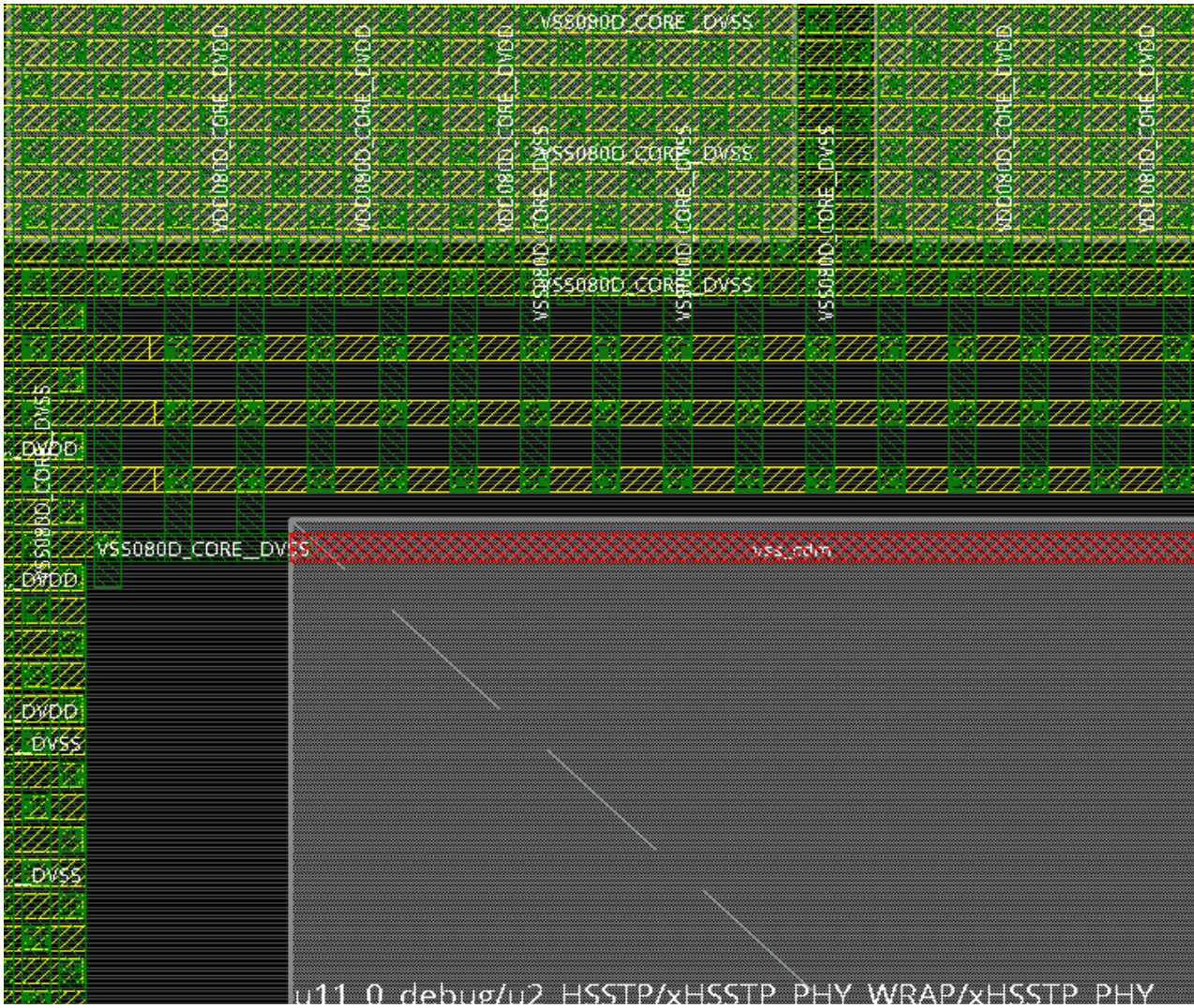
Bumpmap & RDL





Back to back diode (Anti-parallel diode) & ESD

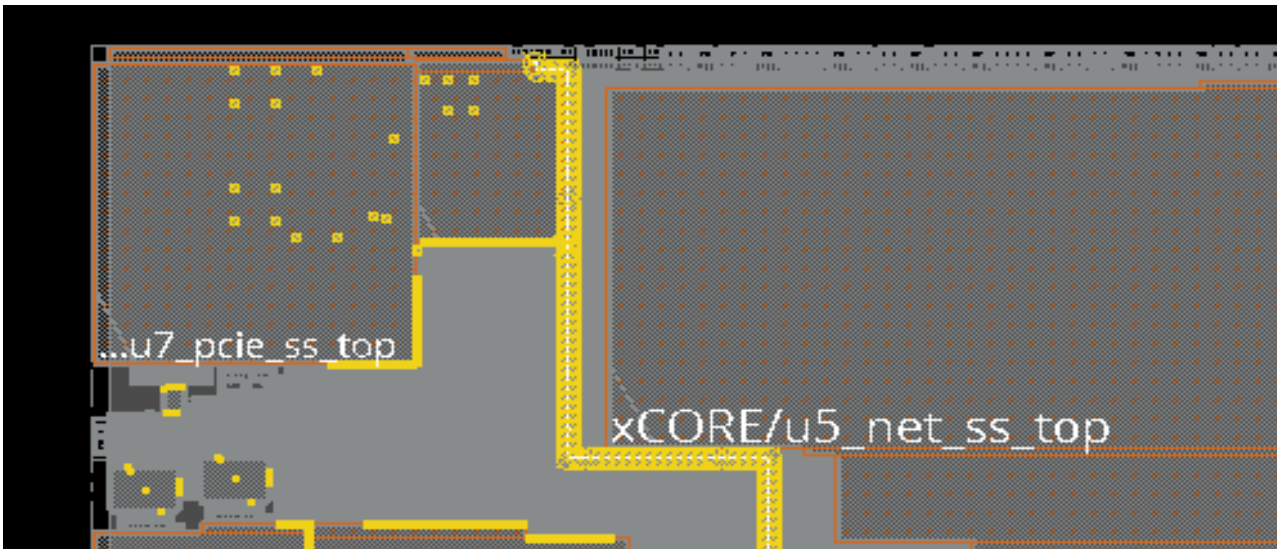


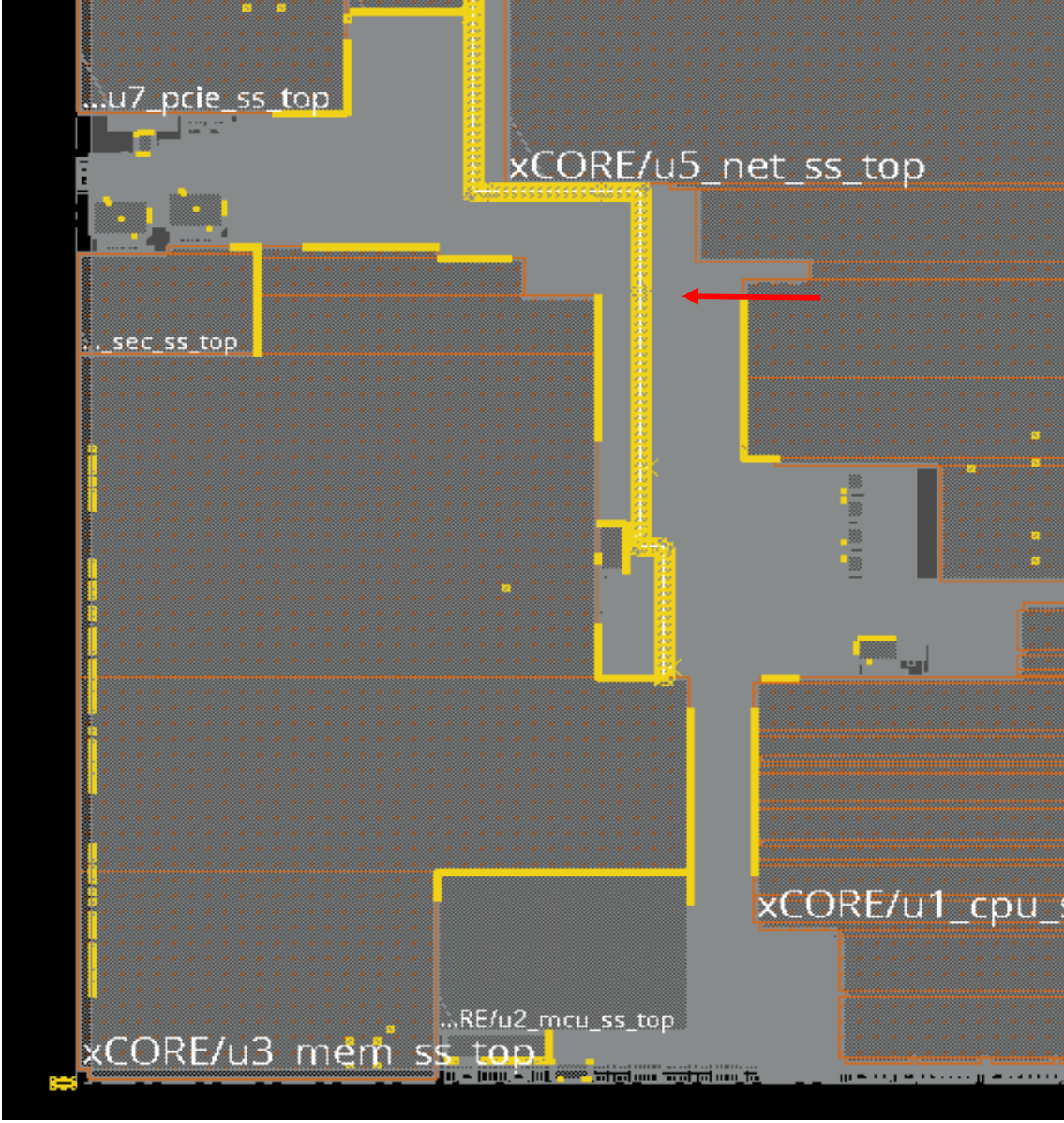


Important signal shielding

OSC Fout to HSSTP "refclk_xtal , refclk_soc_pll"

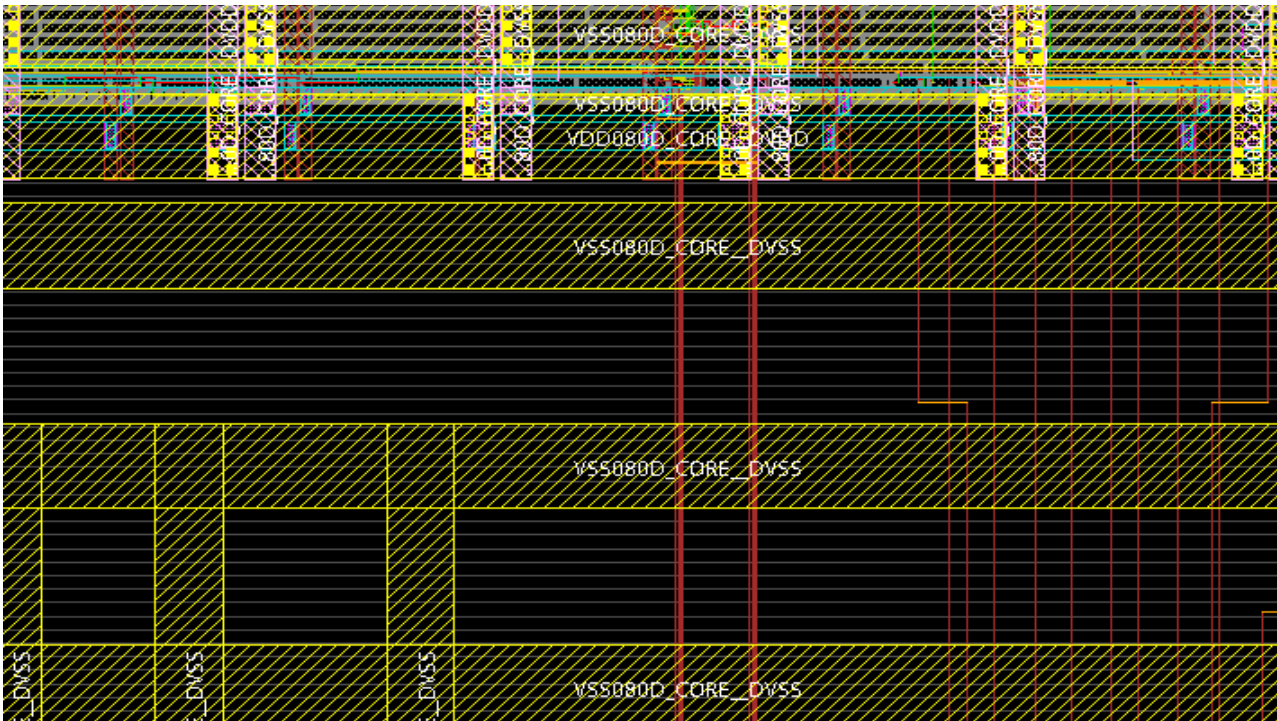
-Top to Sub

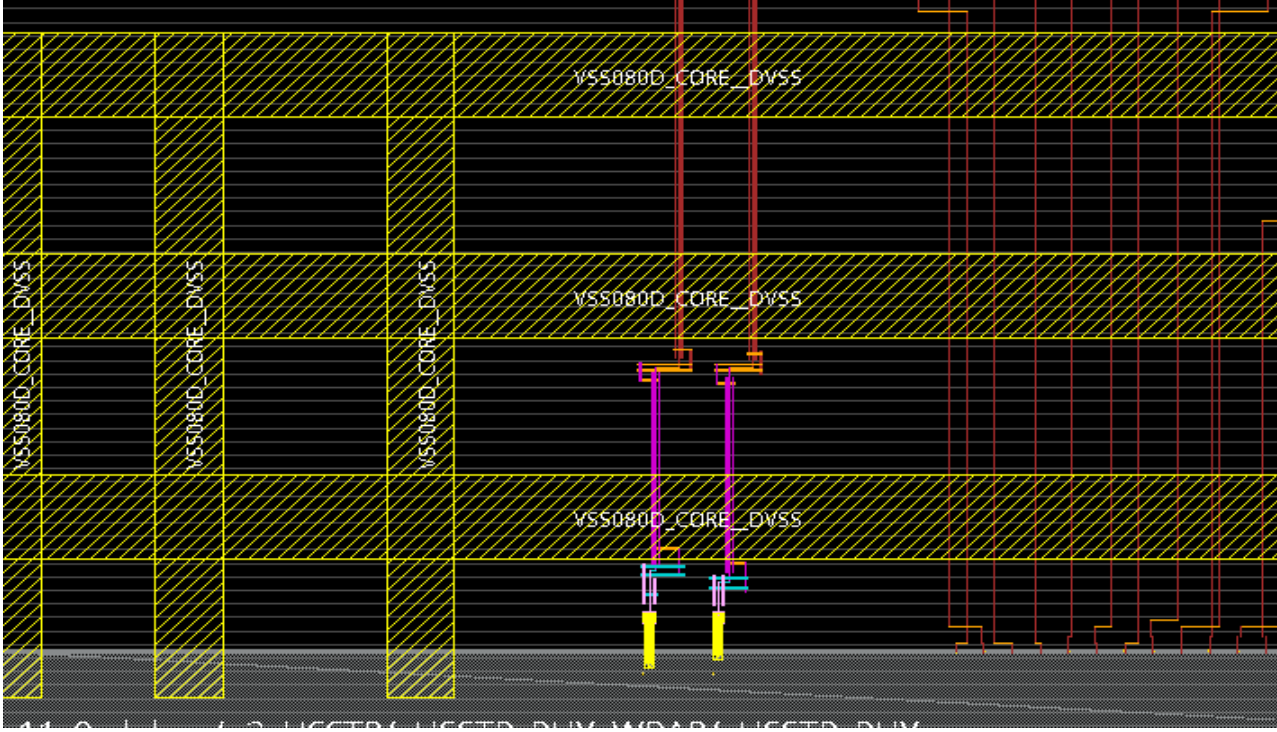




Shielding

-all layer





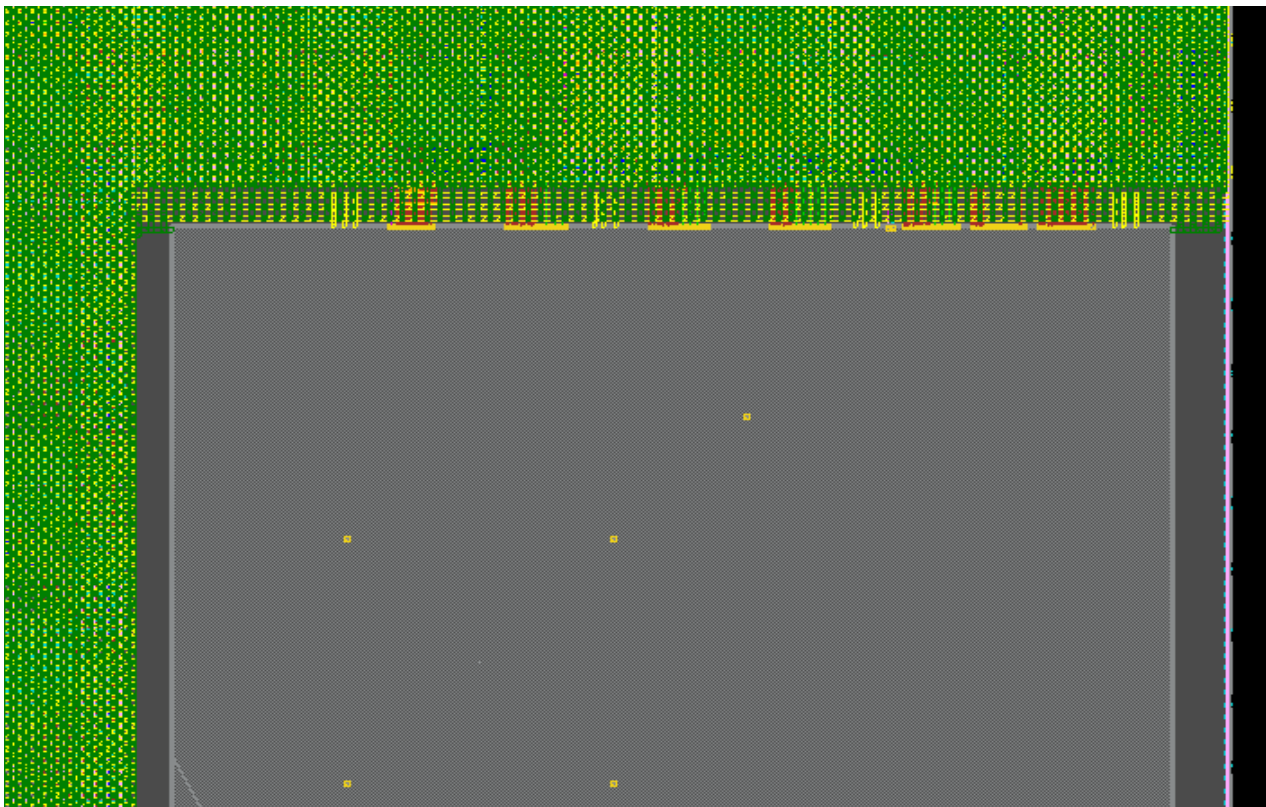
Verification rule check

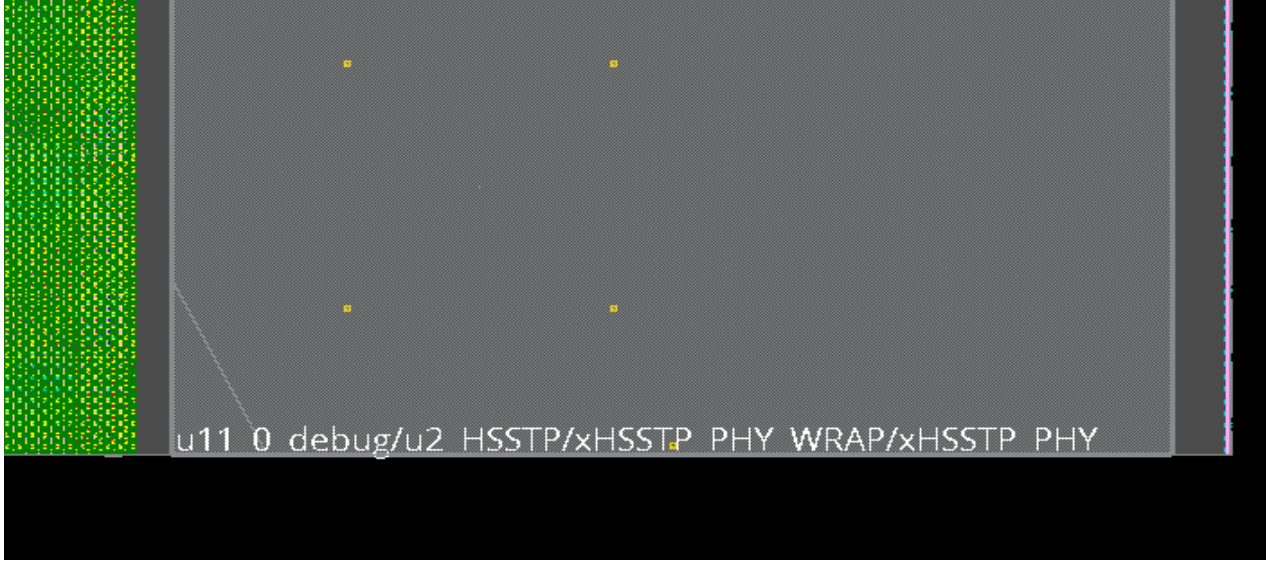
- ERC Result

NO	RULE	
1	ERC_IS18_28	No Path should exist bet

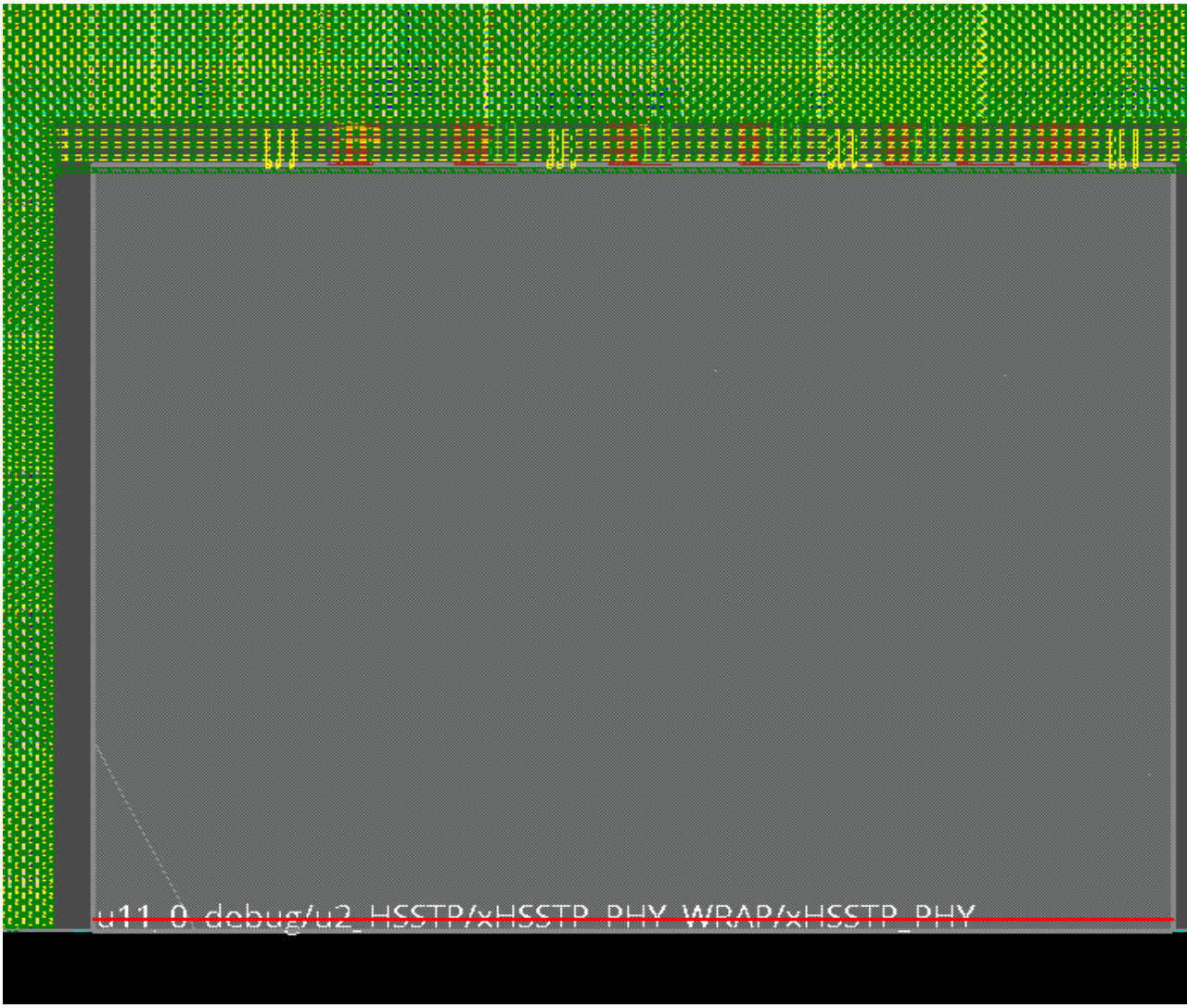
Power & Ground routing

-all Layer

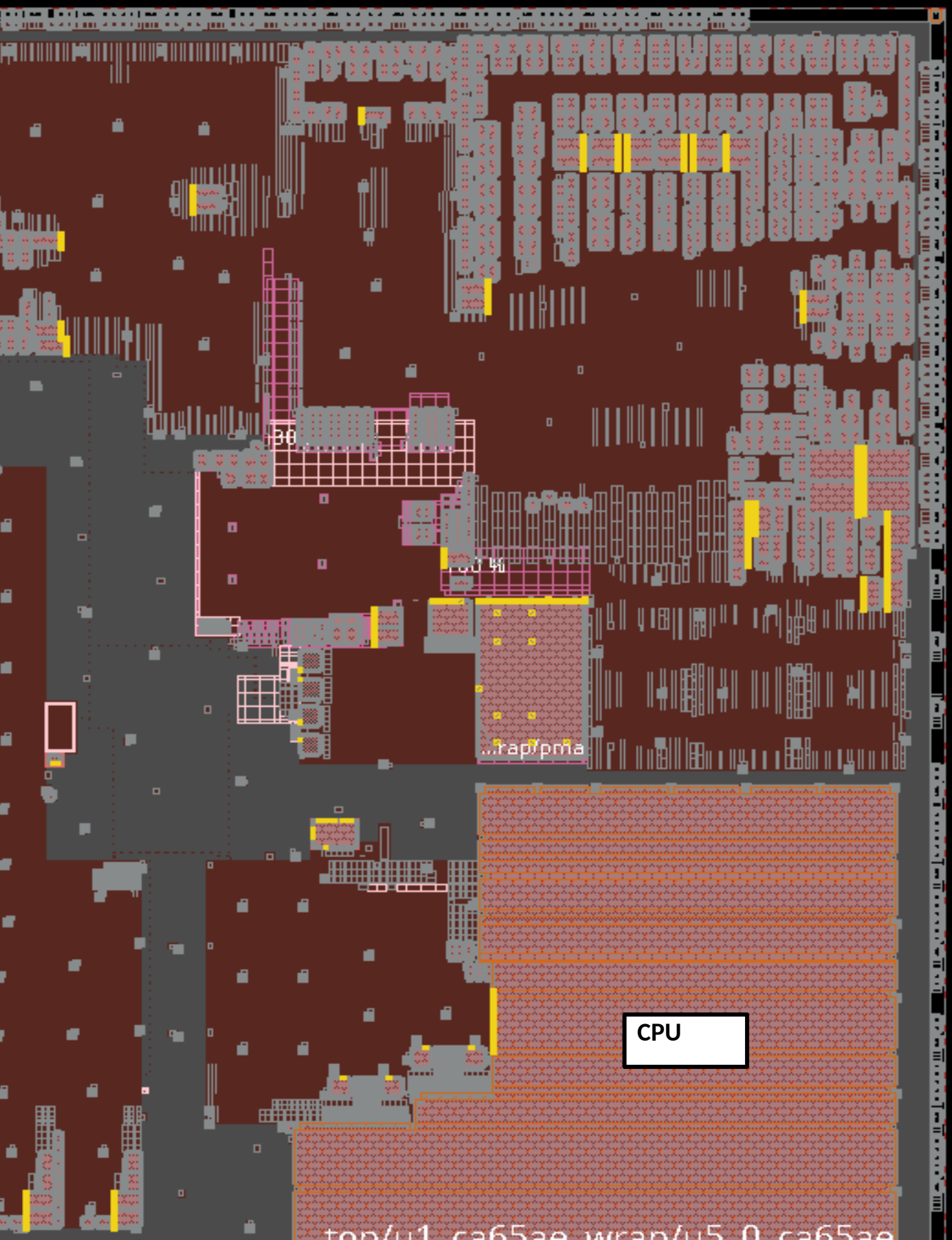




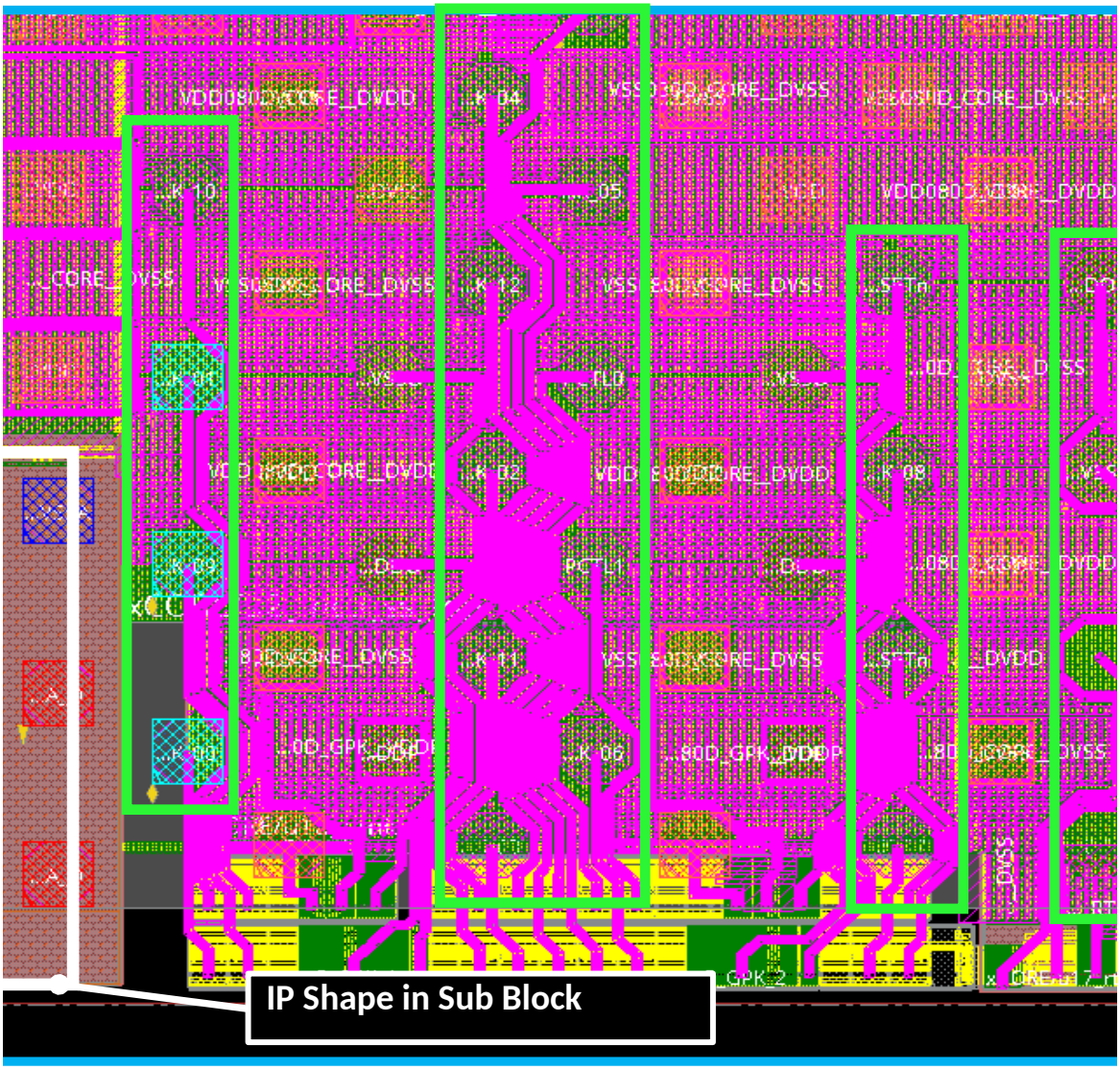
ATB (Analog Test Bus)

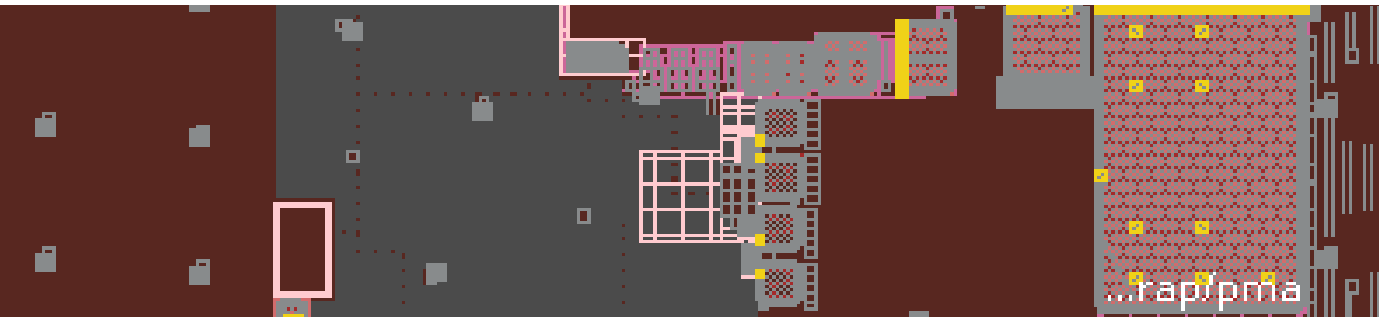


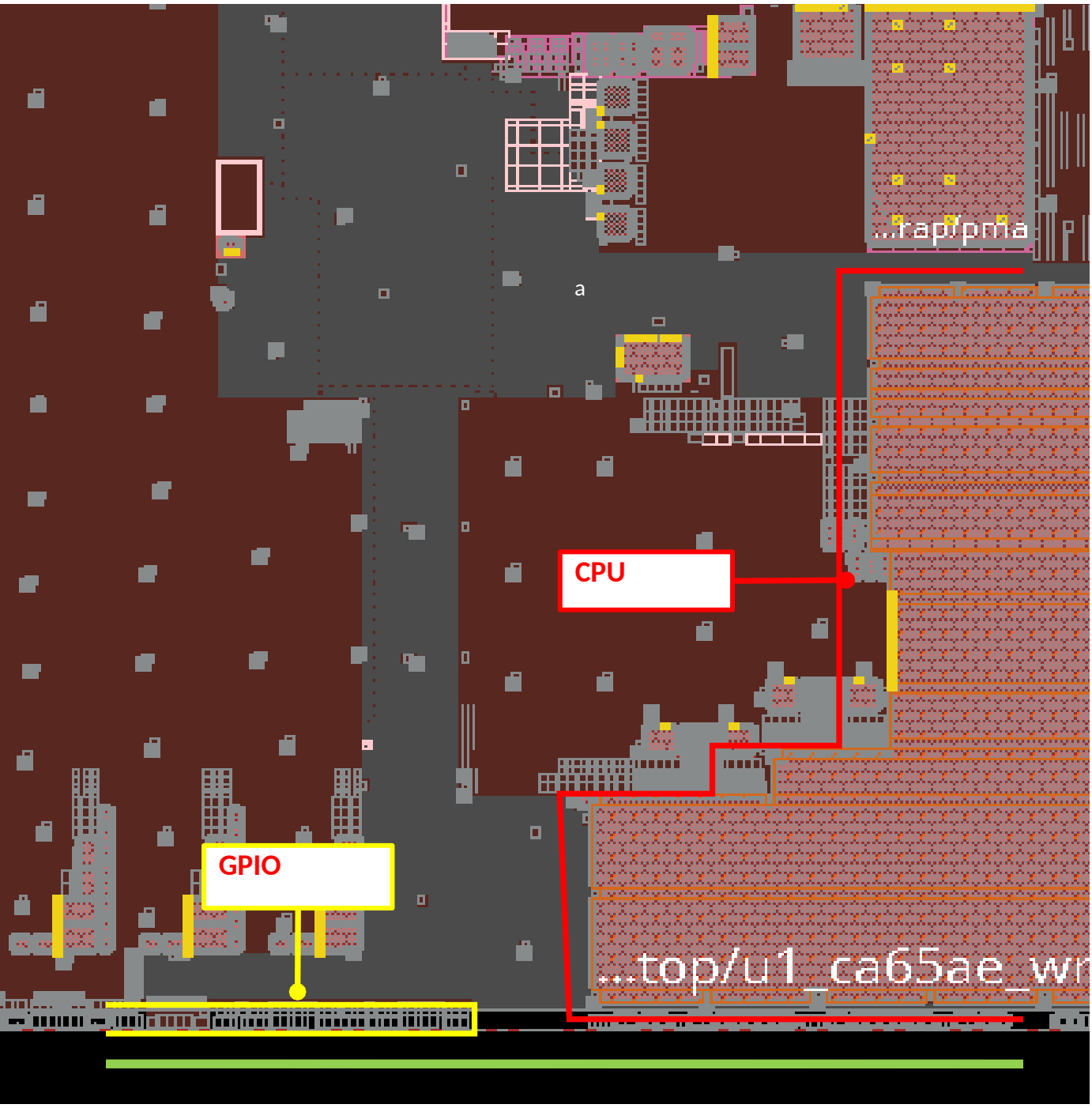
FLOOR PLA



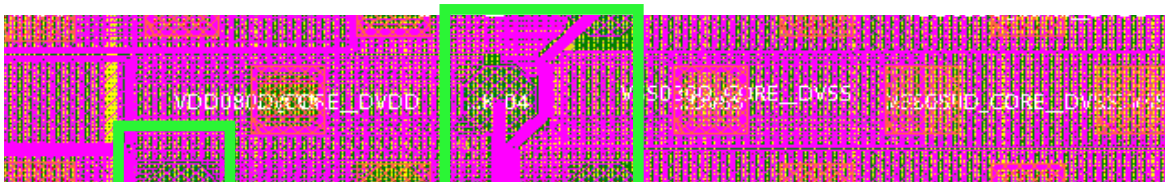
...top/u1_ca65ae_wrap/u5_0_ca65ae

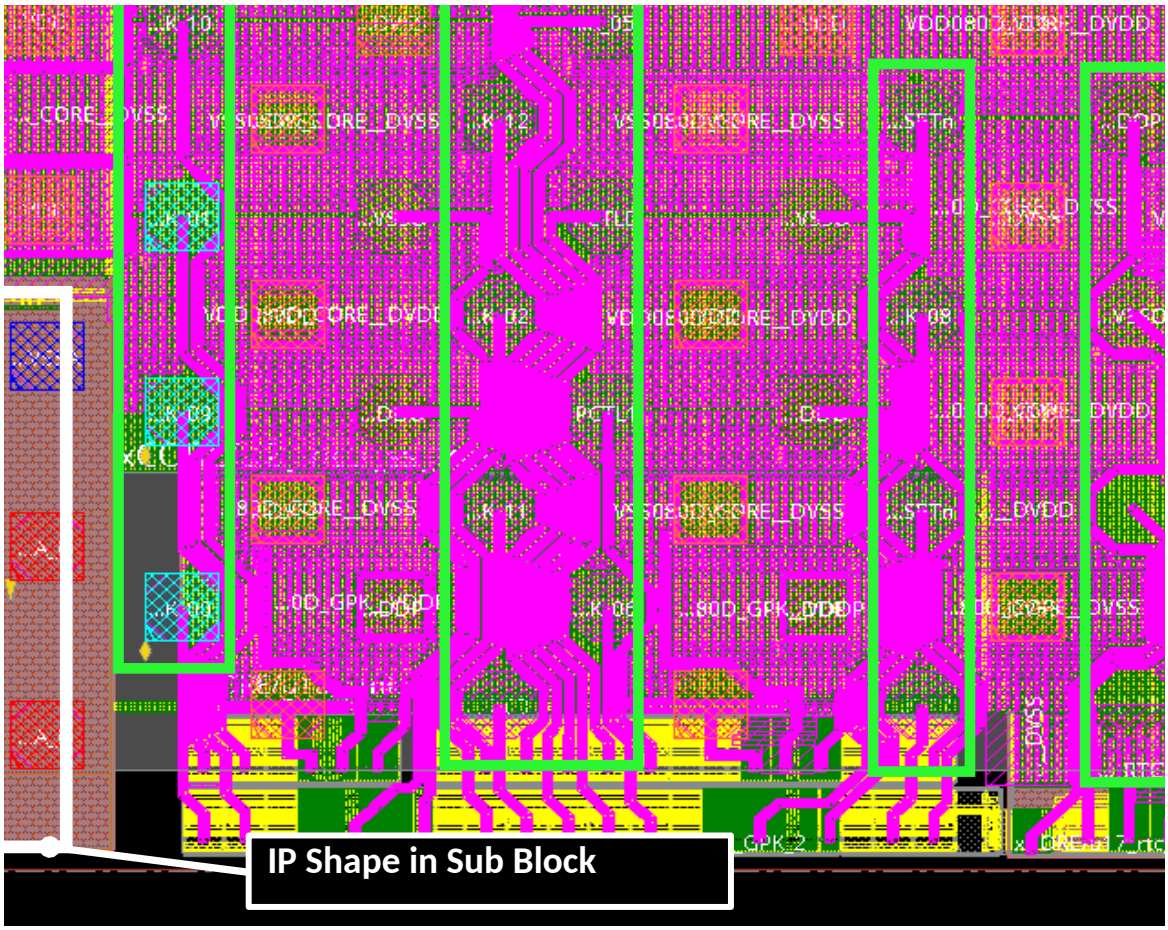

IP Shape in Sub Block

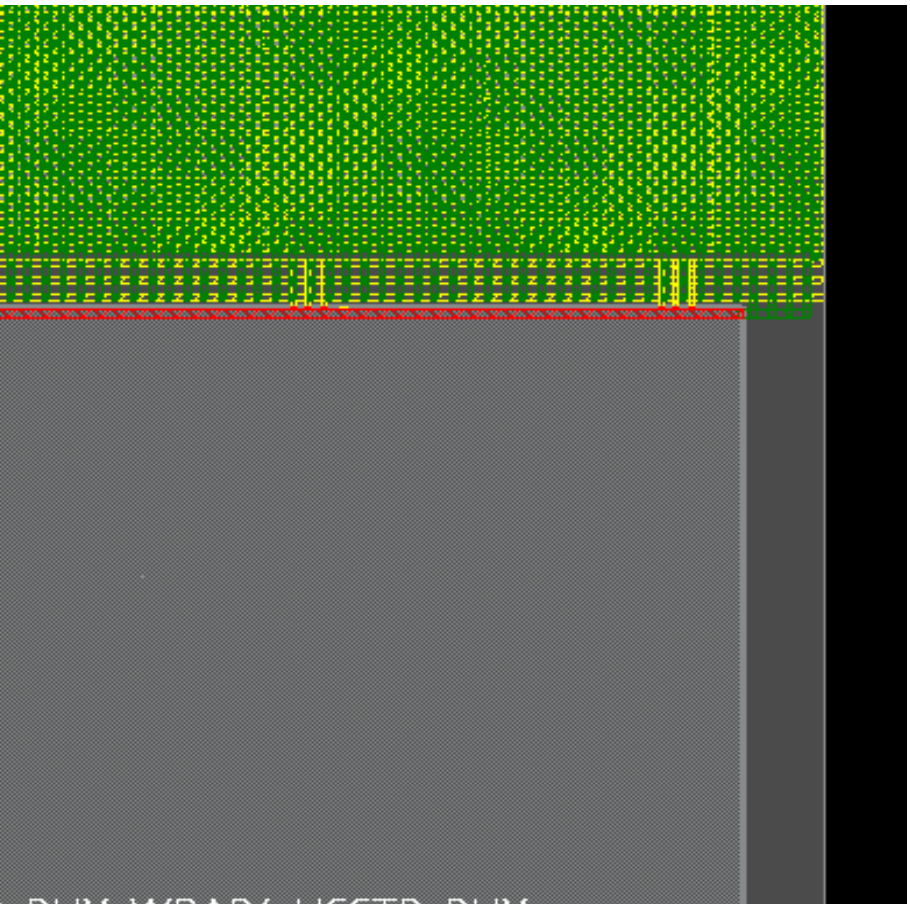

...rap/pma



Integration ch





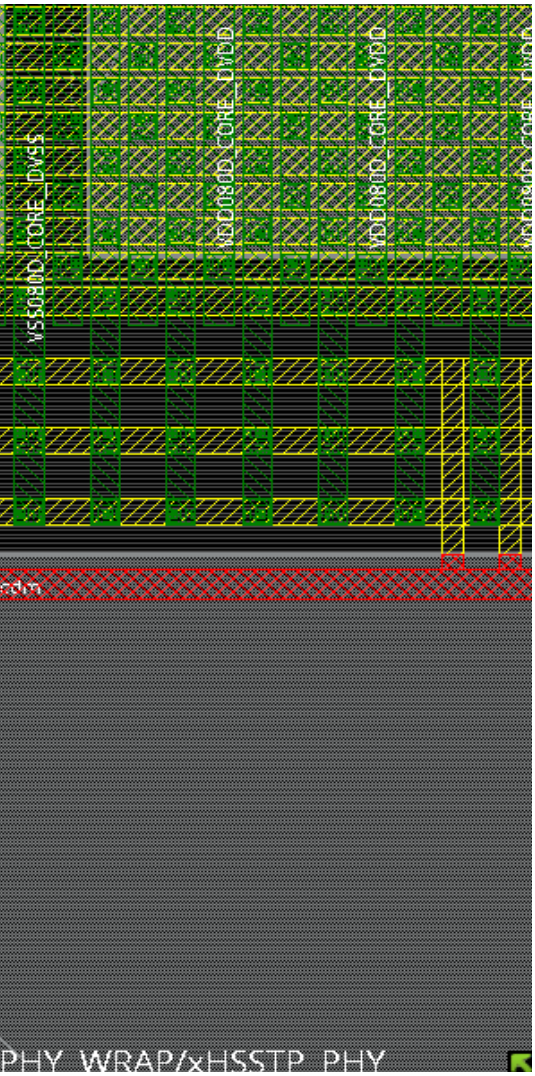


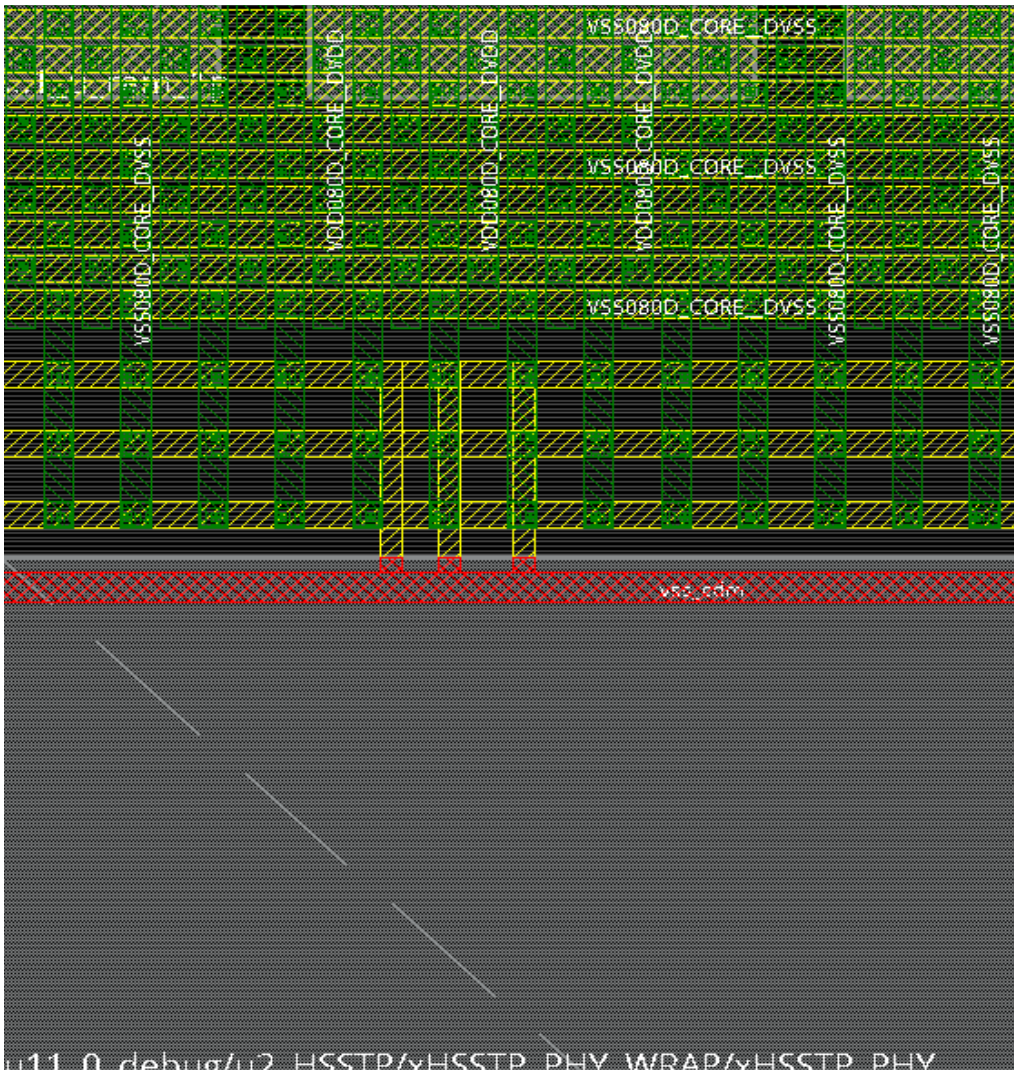
POWER

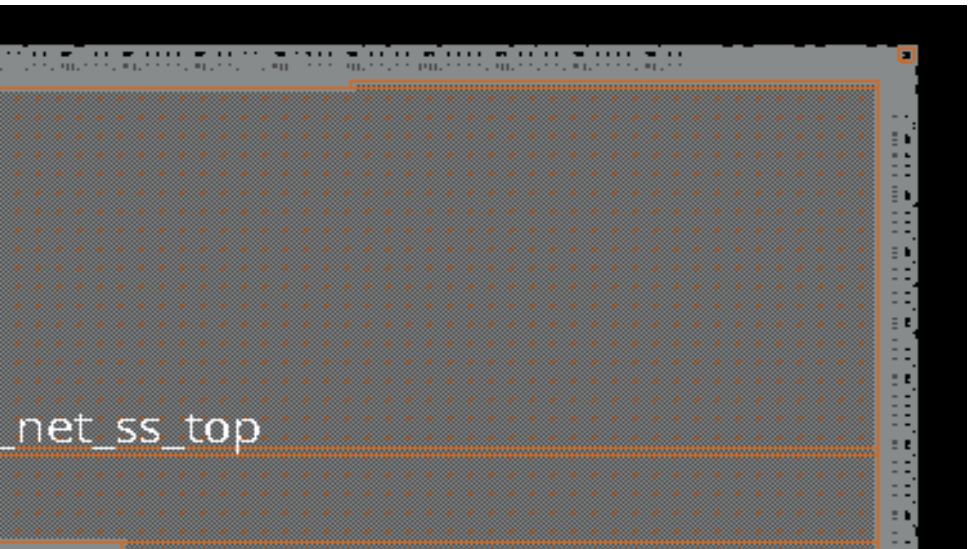
HSSTP vss_cdr

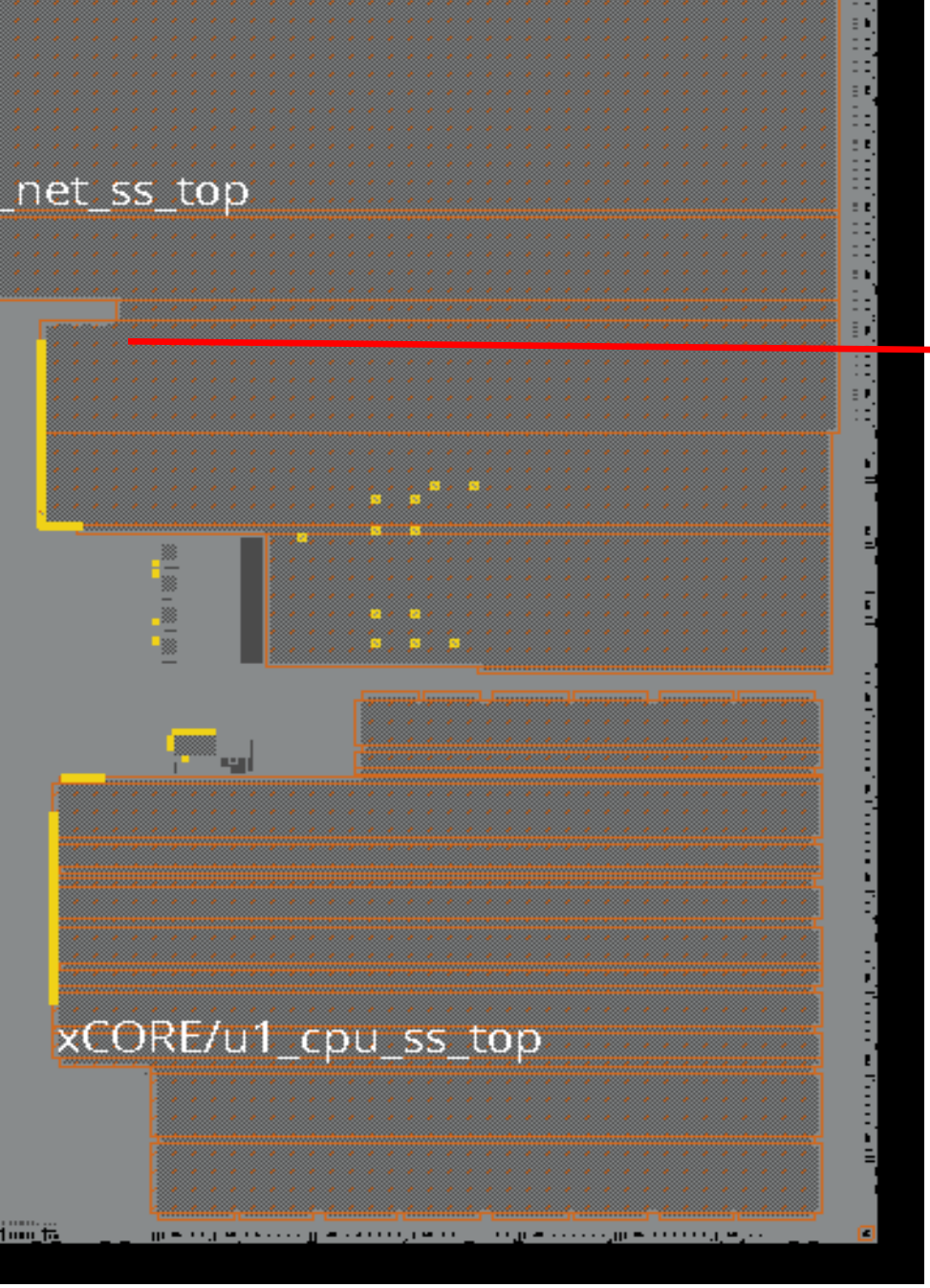
CORE Pwr/Gnd

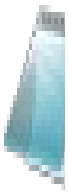
CORE Pwr/Gnd


VSS080D_CORE_DVSS
VDD080D_CORE_DVDD
PHY WRAP/xHSSTP PHY

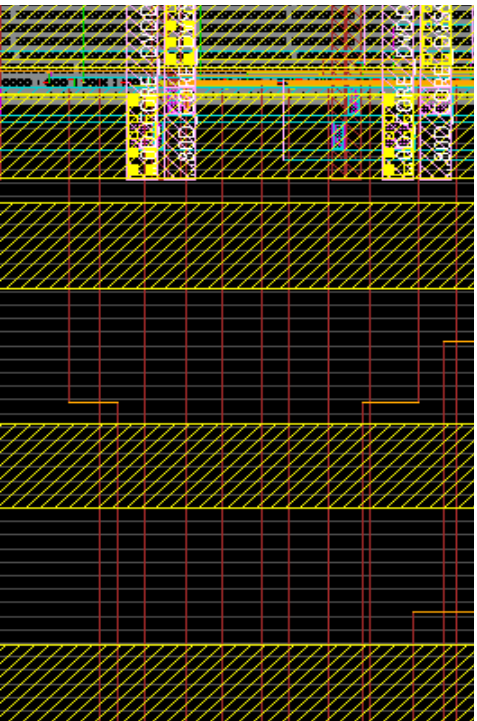

VSS080D_CORE_DVSS
VDD080D_CORE_DVDD
vss_cdm


net_ss_top

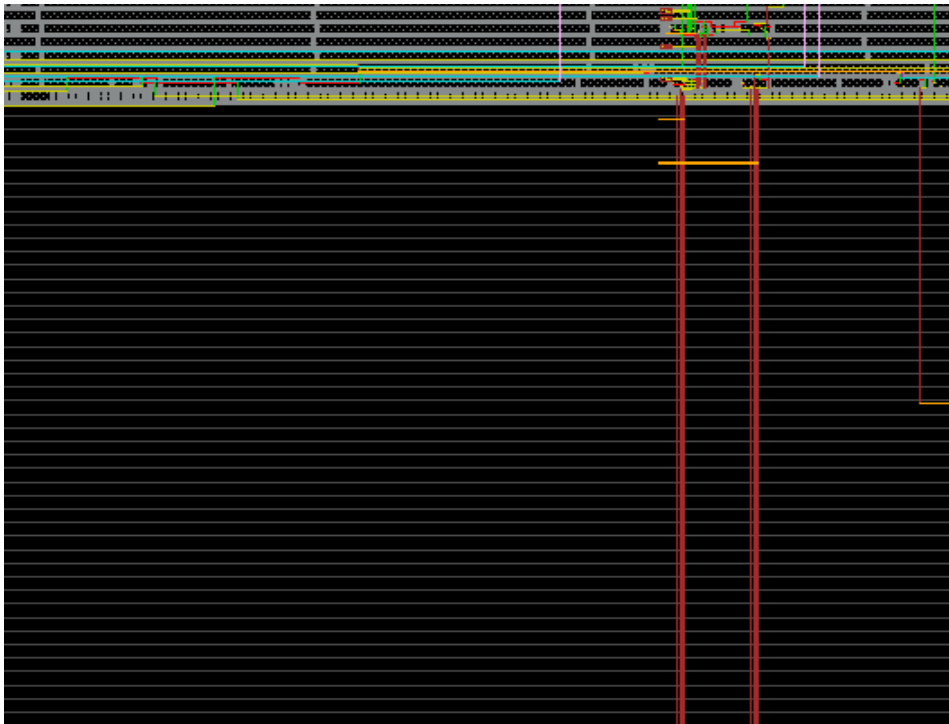


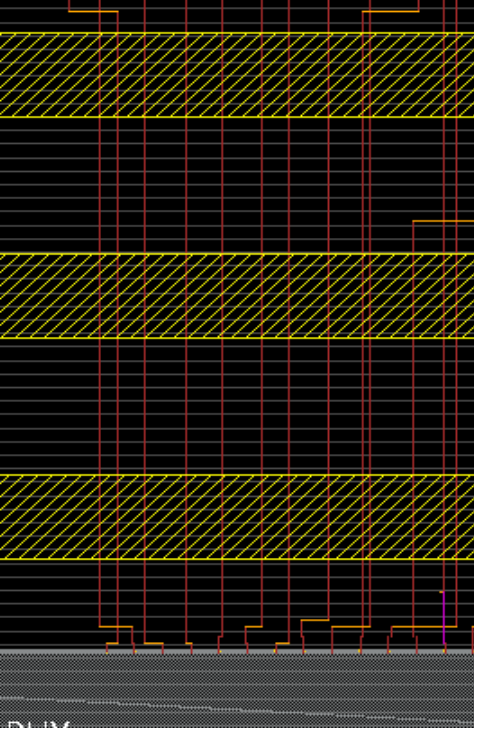


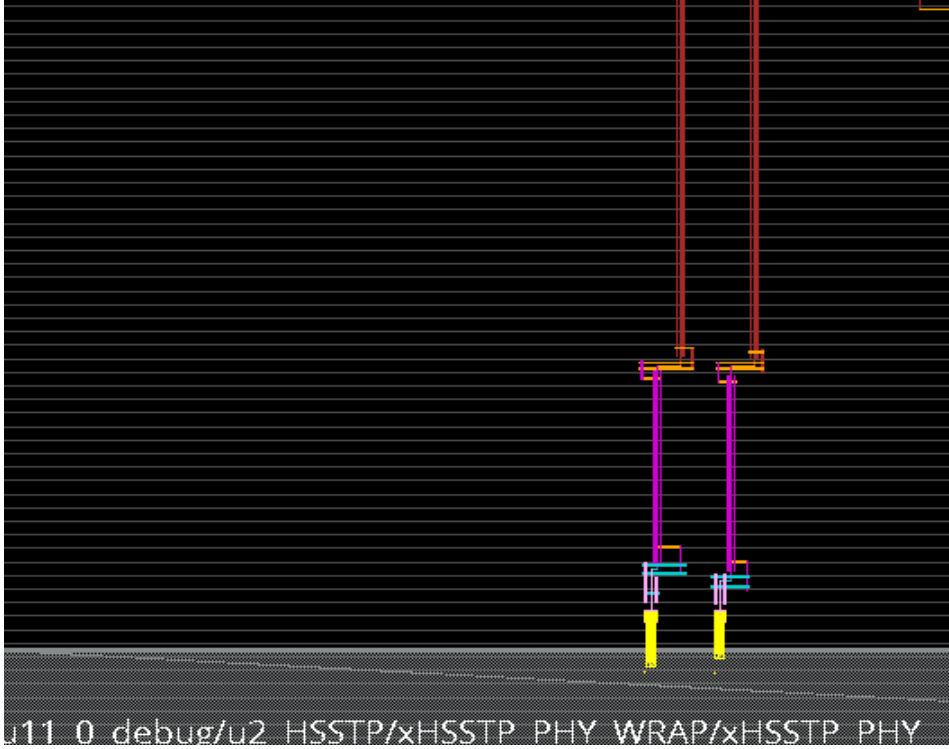
IR_TOP_HSST



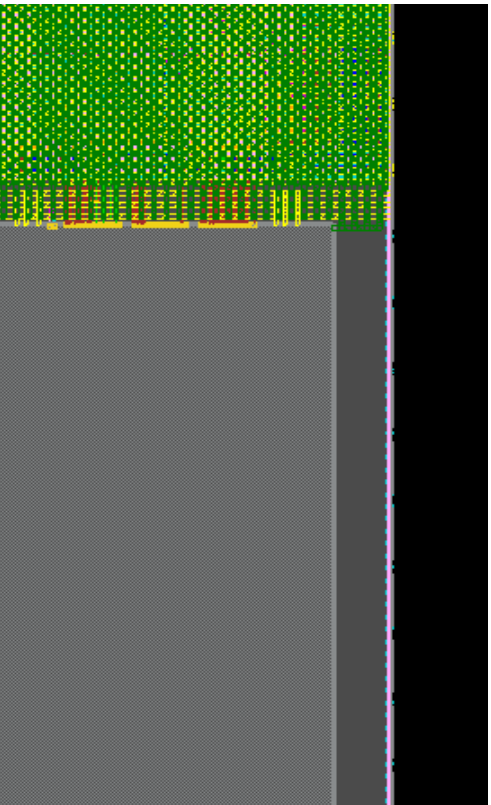
-with out p/g layer

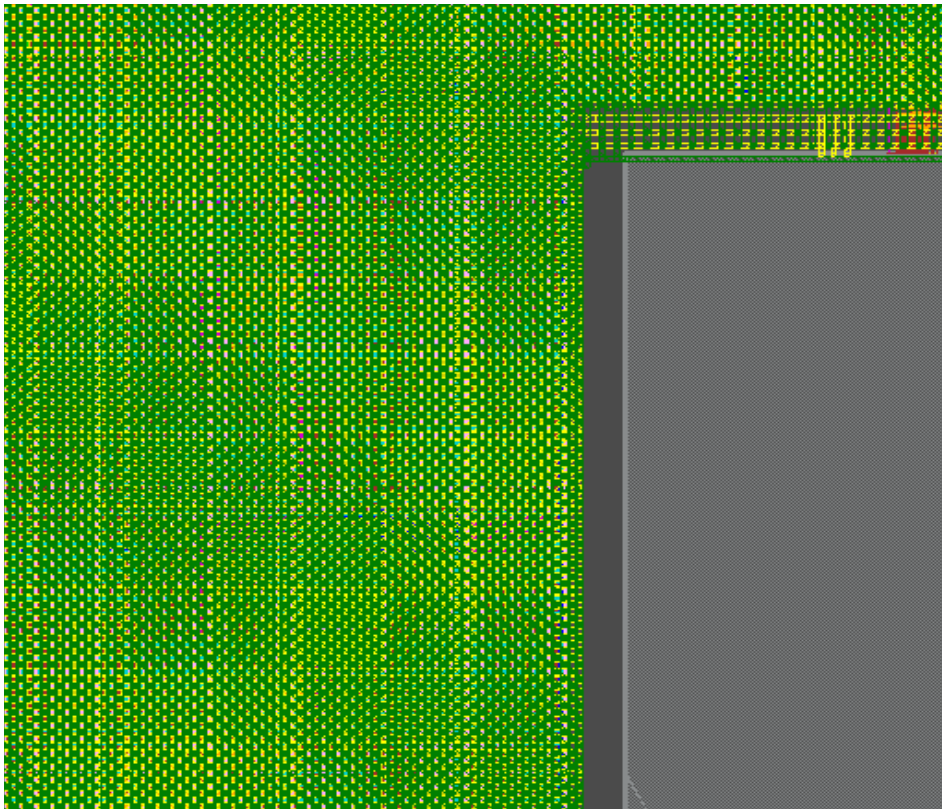


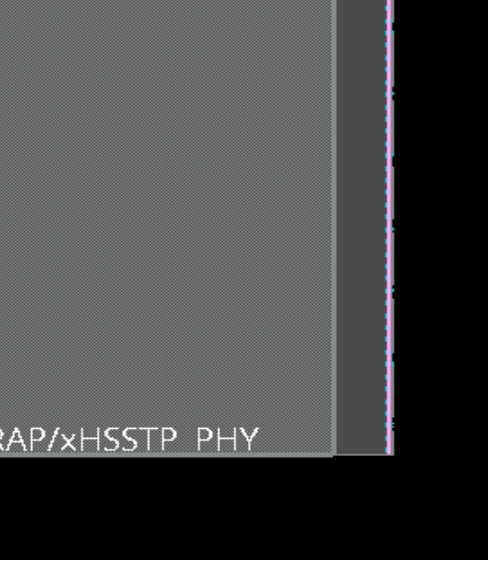


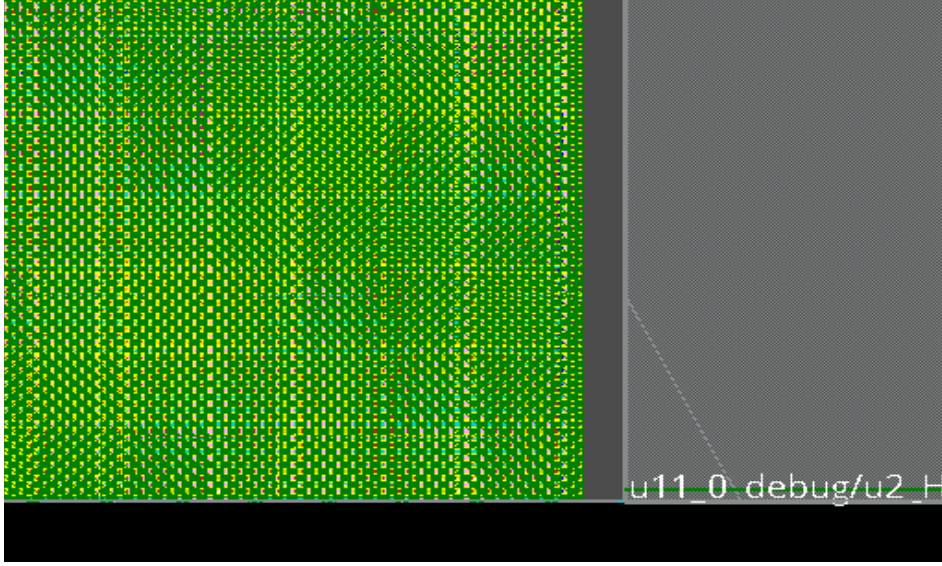


Description	Number of Error	
ween the source/drain of an egpfet and the gate of a thin-oxide MOSFET	1181	I





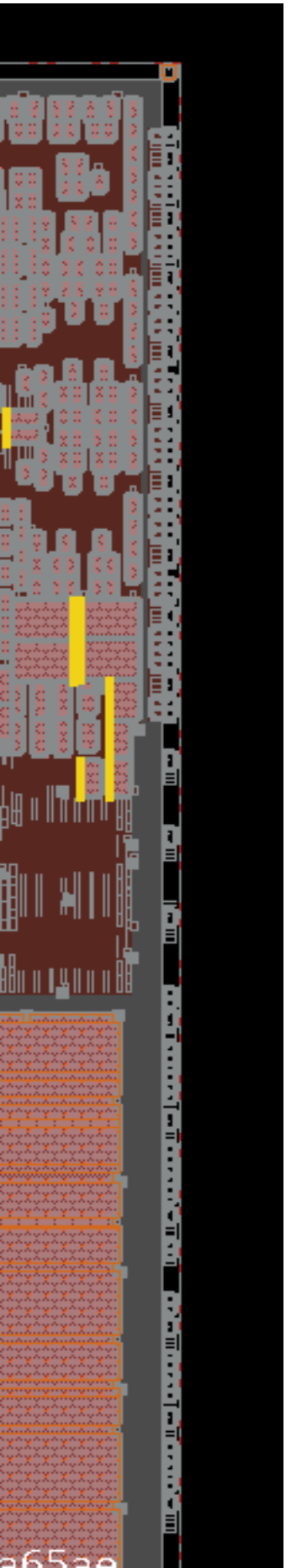

RAP/xHSSTP_PHY

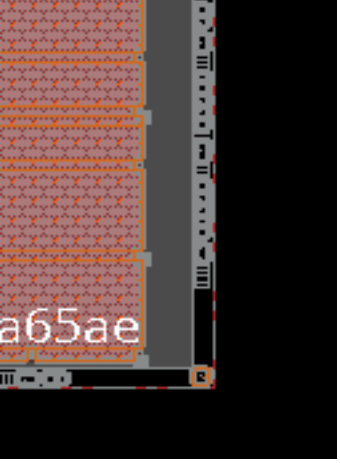

u11_0_debug/u2_H

xHSSTP_PHY

VSS080D_CORE DVSS
VDD080D_CORE DVDD
u11_0

N

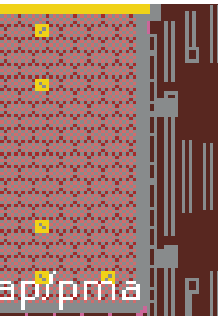




BUMP

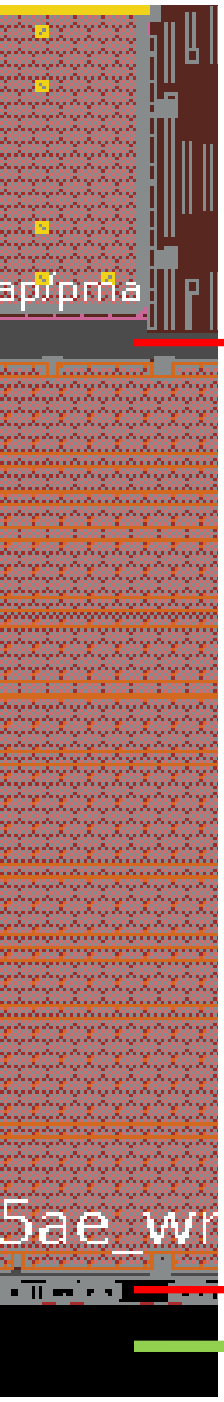
HSSTP_POWER_BUMP
HSSTP_GROUND_BUMP
HSSTP_SIGNAL_BUMP

GPIO_BUMP
CORE_POWER/GROUND BUMP



Distance

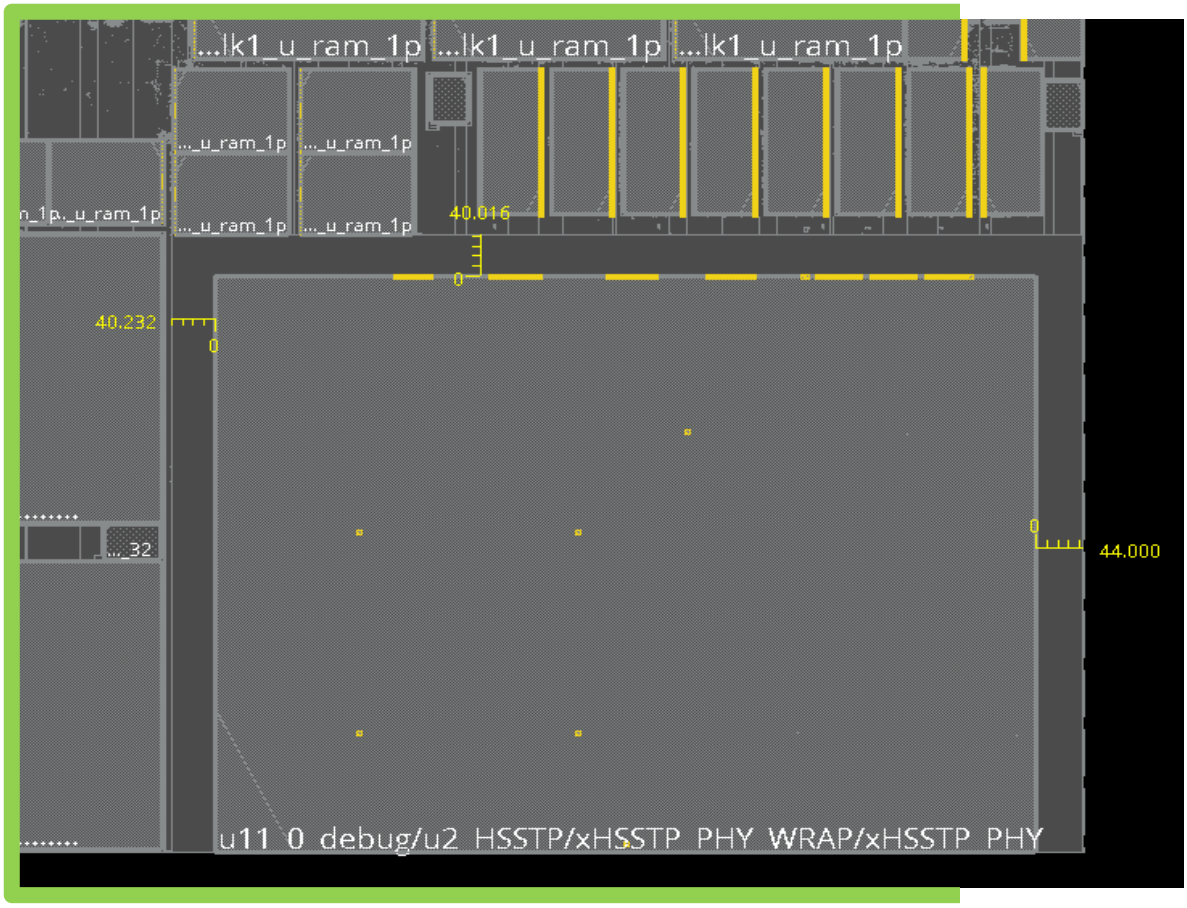
HSSTP to DDR PHY : 1635um



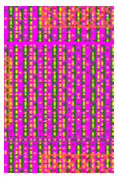
HSSTP to CPU : 2435um
HSSTP to GPIO : 110um

HSSTP to CORE Logic : 40um

From the Chip Edge : 1940um



eck



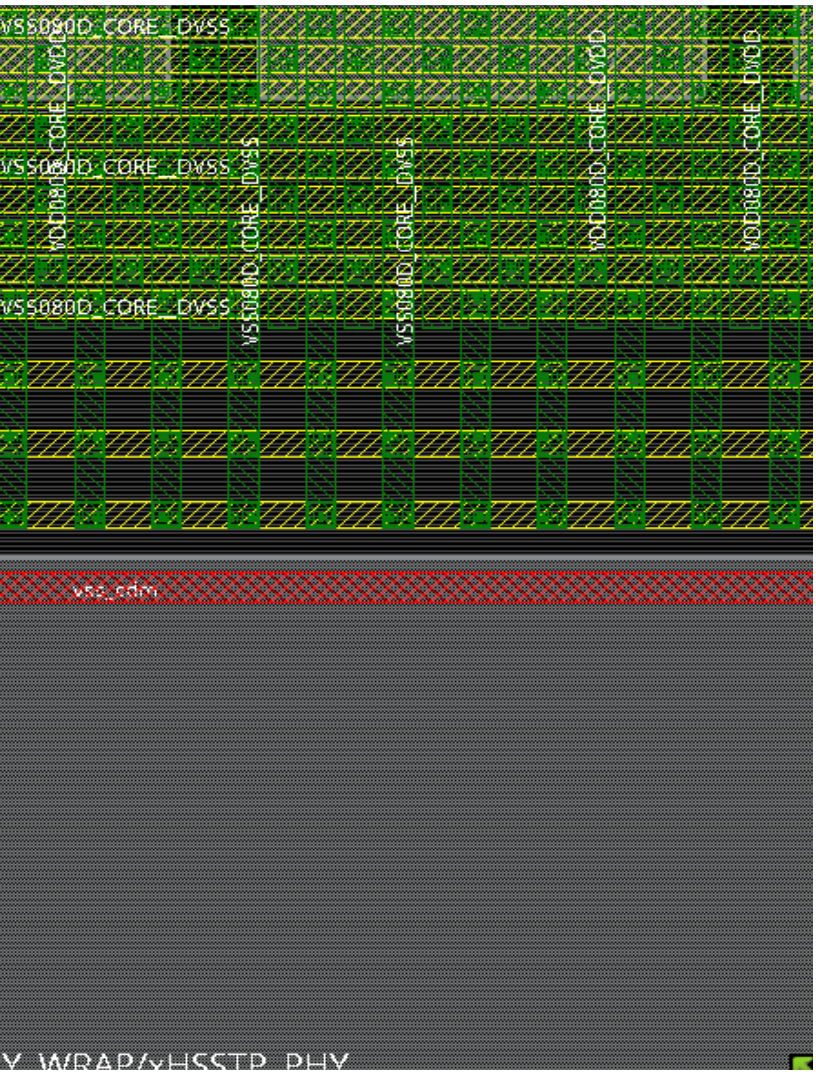
BUMP

HSSTP_POWER_BUMP
HSSTP_GROUND_BUMP
HSSTP_SIGNAL_BUMP

GPIO_BUMP
CORE_POWER/GROUND BUMP

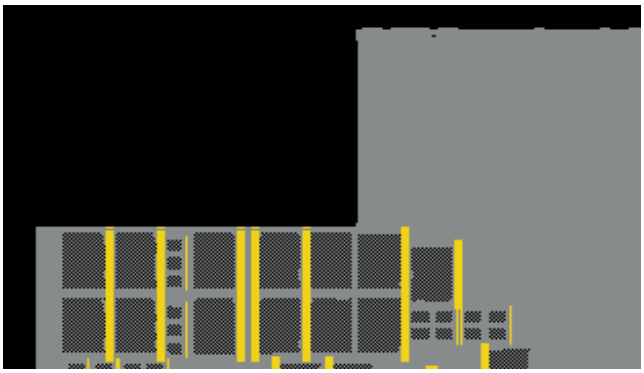
m

d Layer G1
d Layer G2

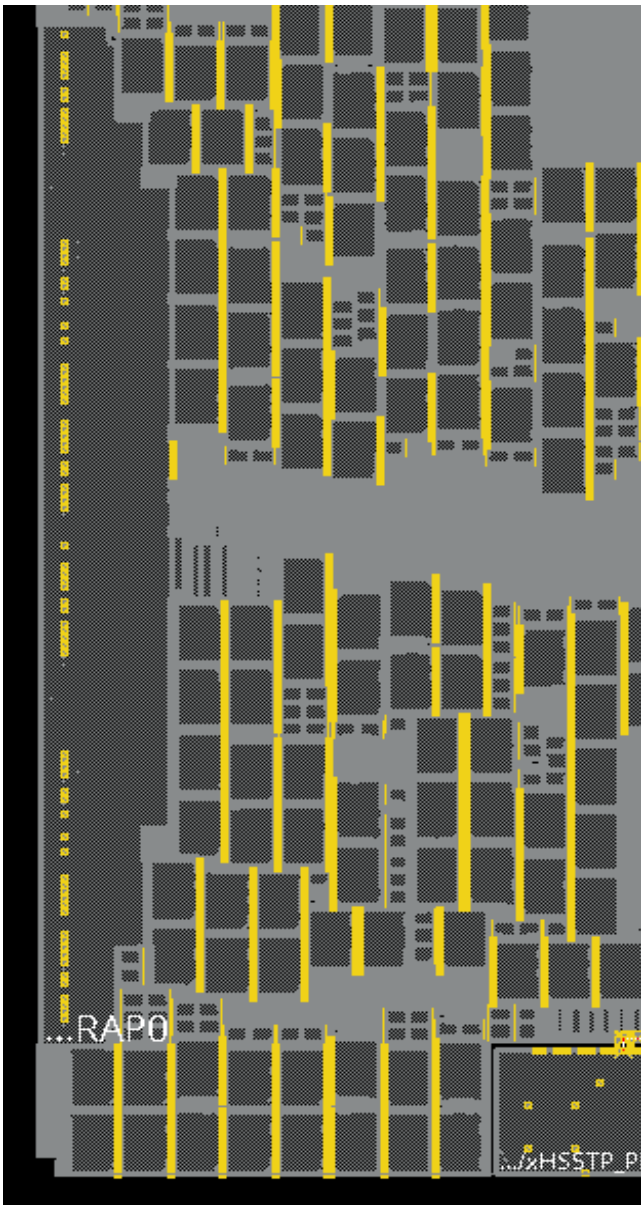

VSS080D_CORE_DVSS
VDD080D_CORE_DVDD
VSS080D_CORE_DVSS
VDD080D_CORE_DVDD
VSS080D_CORE_DVSS
VSS080D_CORE_DVSS
VSS080D_CORE_DVSS
VDD080D_CORE_DVDD
VDD080D_CORE_DVDD
vss_cdm
Y WRAP/xHSSTP PHY

VSS080D_CORE
VSS080D_CORE
VSS080D_CORE
VSS080D_CORE_DVSS
VDD080D_CORE_DVDD
VDD080D_CORE_DVDD
VDD080D_CORE_DVDD
VSS080D_CORE_DVSS
vss_cdm
u11_0_debug/u2_HSSTP/xHSSTP_PHY_WRAP/

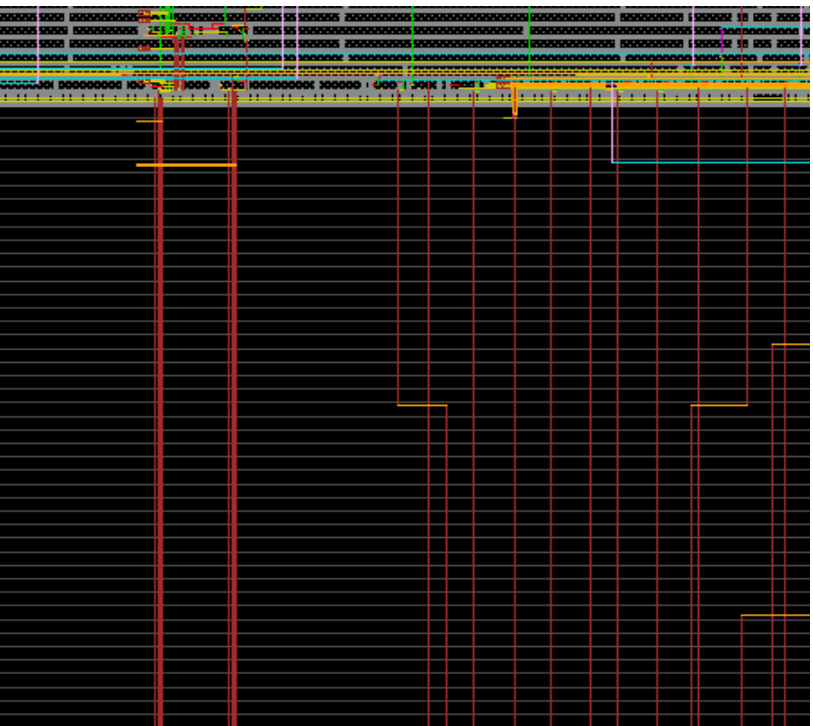
-Sub to IP

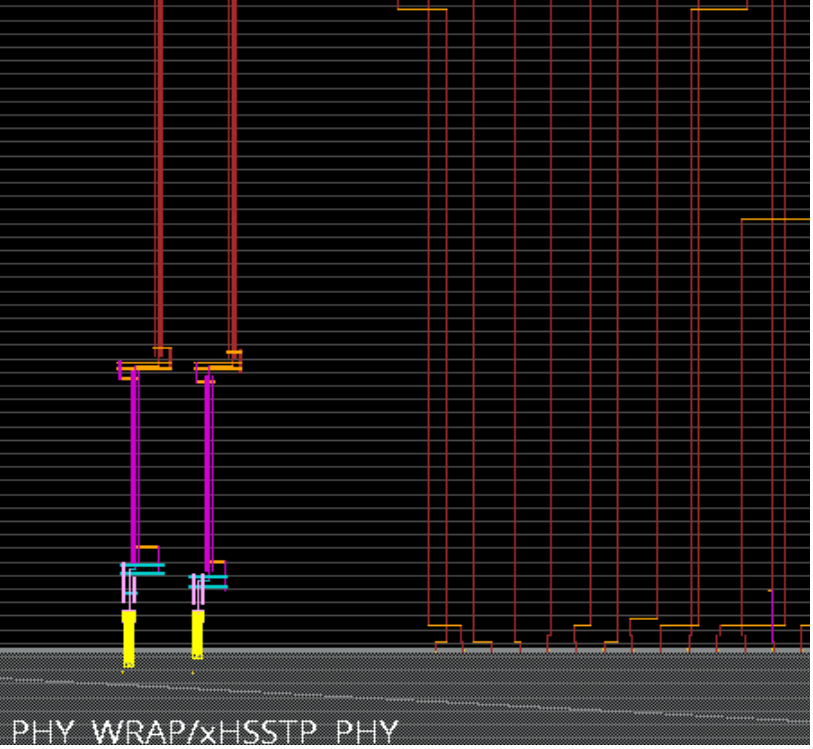


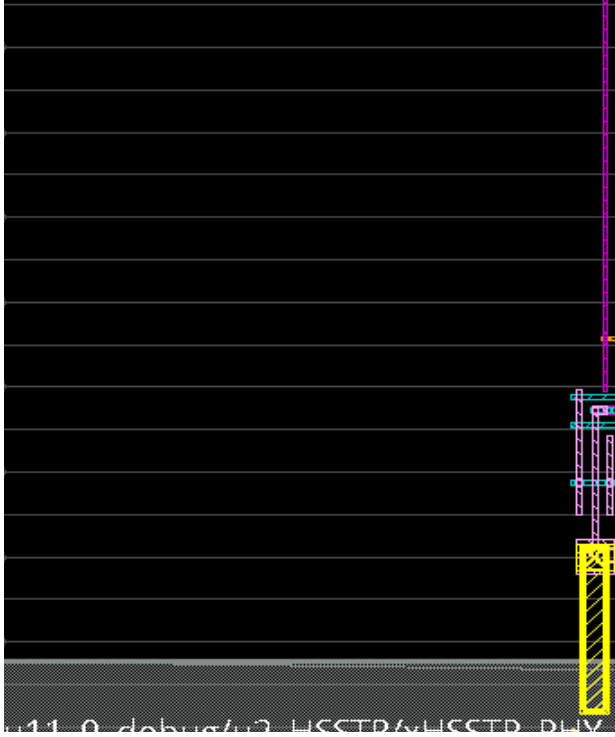
P_shield.rpt



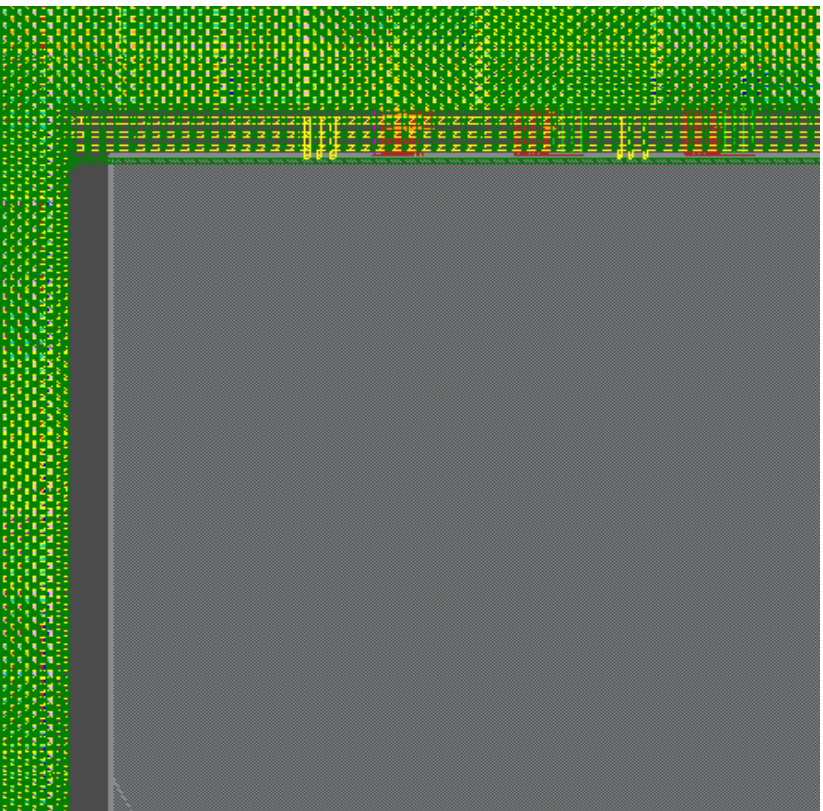
- zoom to Signal



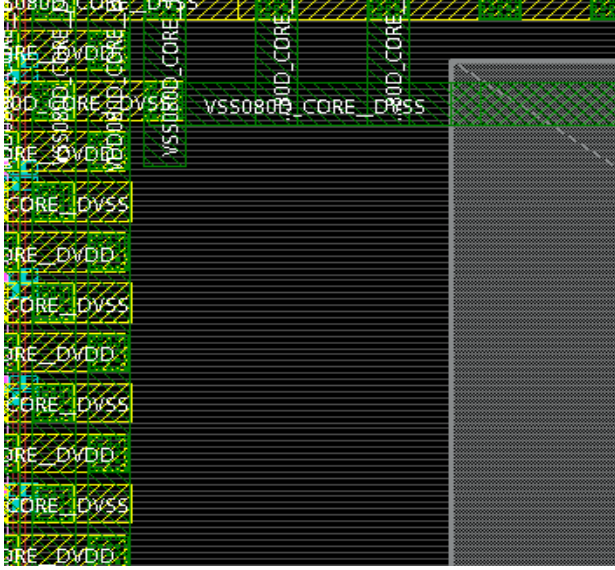




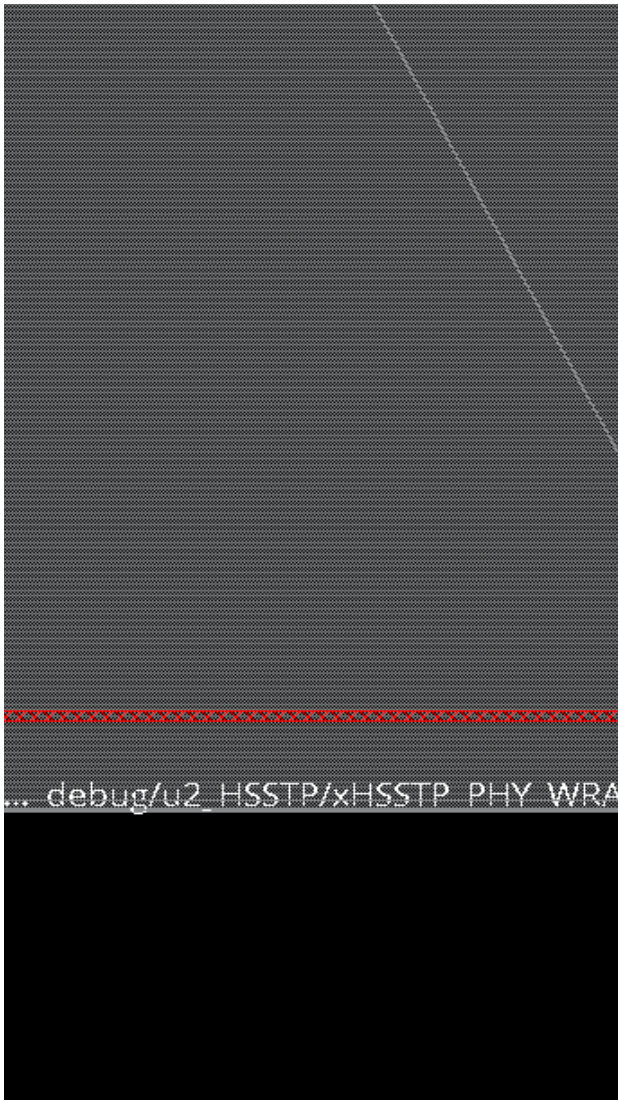
Comments	Location
Exists in results of the received PV	rs_hsstptx2ch_ln14lpp_v1p0

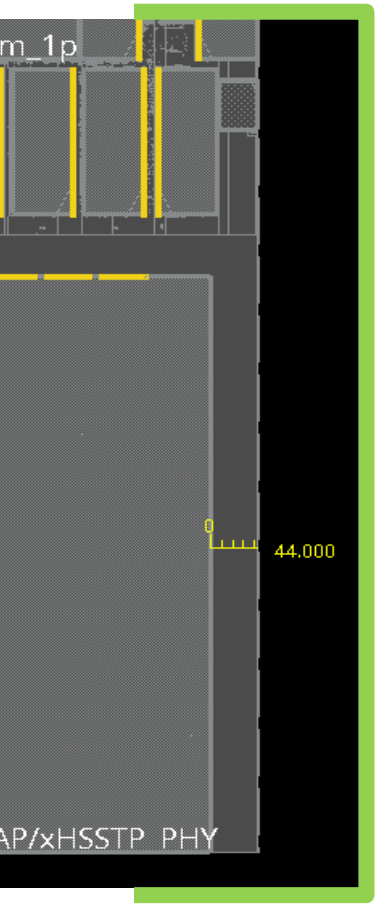


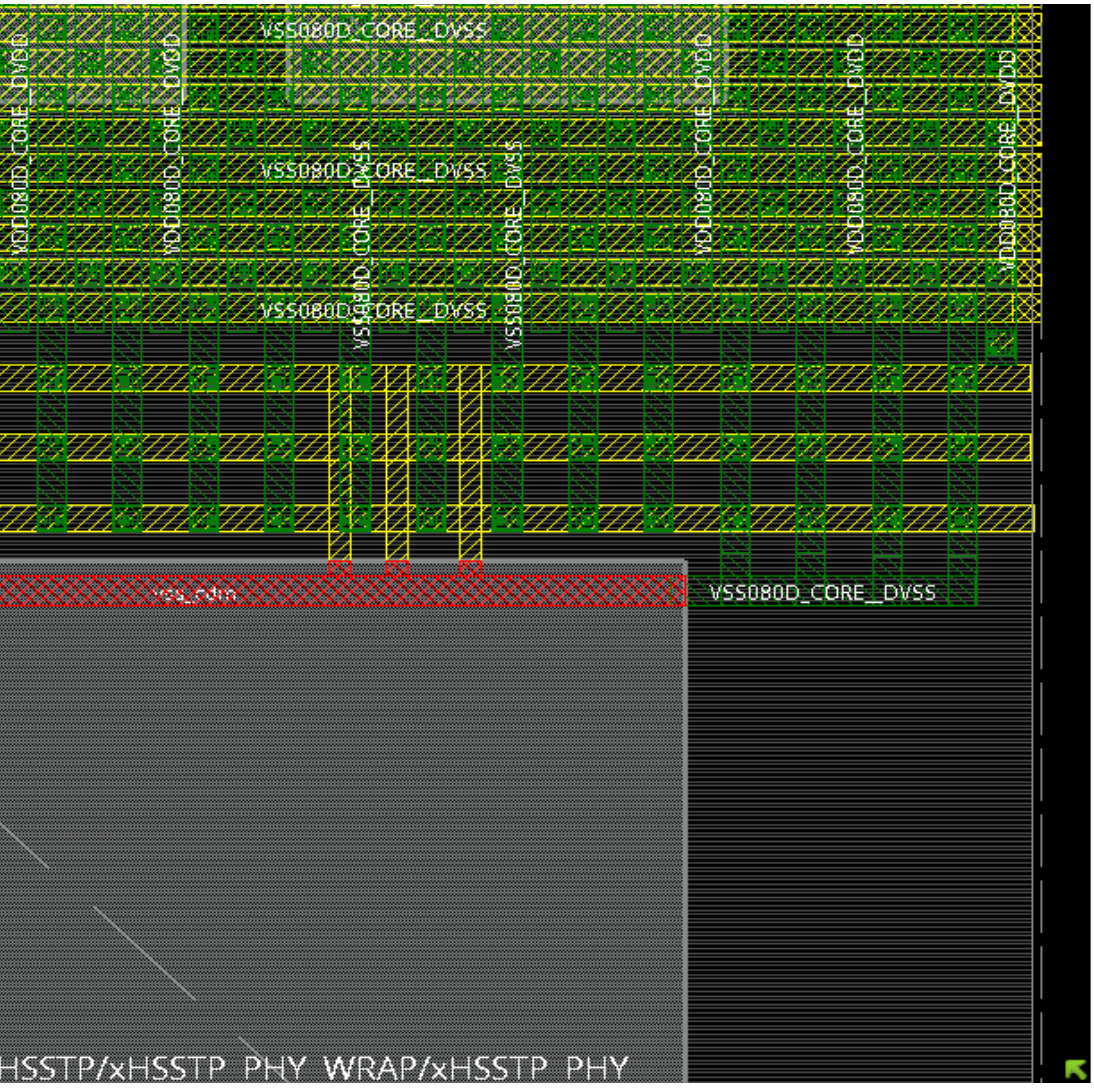
u11_0_debug/u2_HSSTP/xHSSTP_PHY_WRA

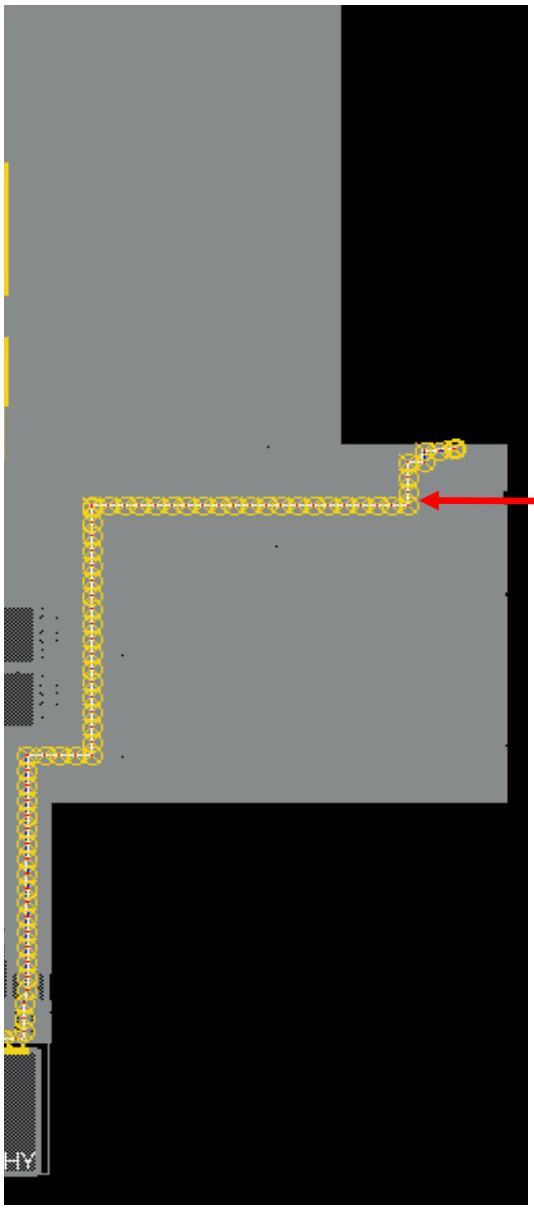

VSS0800_CORE_DVSS

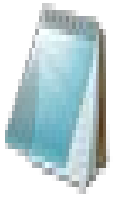
VDD0800_CORE_DVDD
u11_0_debug/u2_HSSTP/xHSSTP


... debug/u2_HSSTP/xHSSTP_PHY_WRA

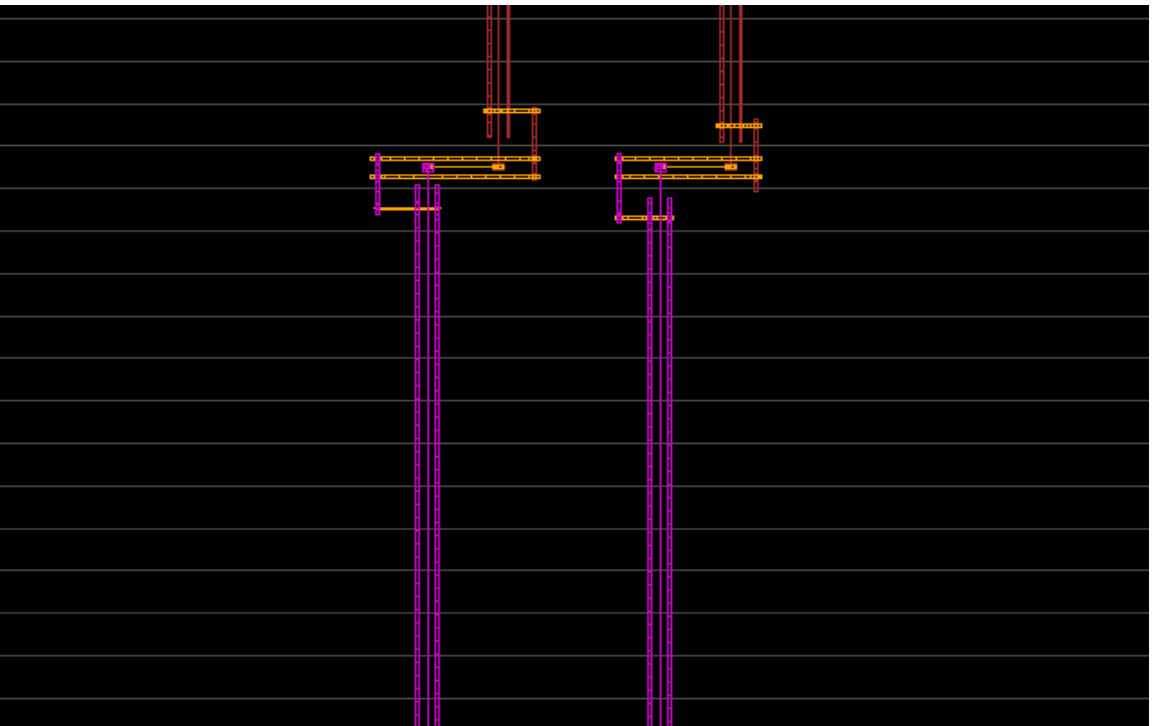

m_1p
0
44.000
AP/xHSSTP_PHY

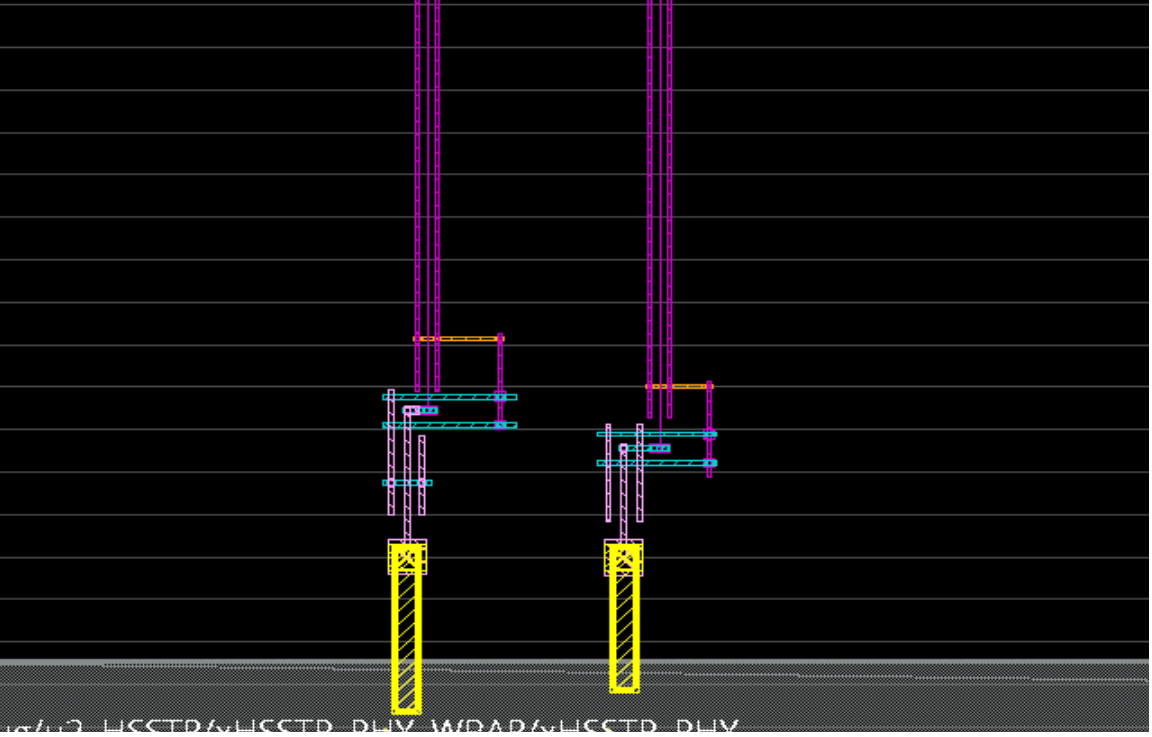

VSS080D_CORE_DVSS
VDD080D_CORE_DVDD
VSS080D_CORE_DVSS
VDD080D_CORE_DVDD
VSS080D_CORE_DVSS
VDD080D_CORE_DVDD
VSS080D_CORE_DVSS
VDD080D_CORE_DVDD
VSS080D_CORE_DVSS
VDD080D_CORE_DVDD
VSS080D_CORE_DVSS
VSS080D_CORE_DVSS
HSSTP/xHSSTP PHY WRAP/xHSSTP PHY





IR_HSSTP_shielding.rpt

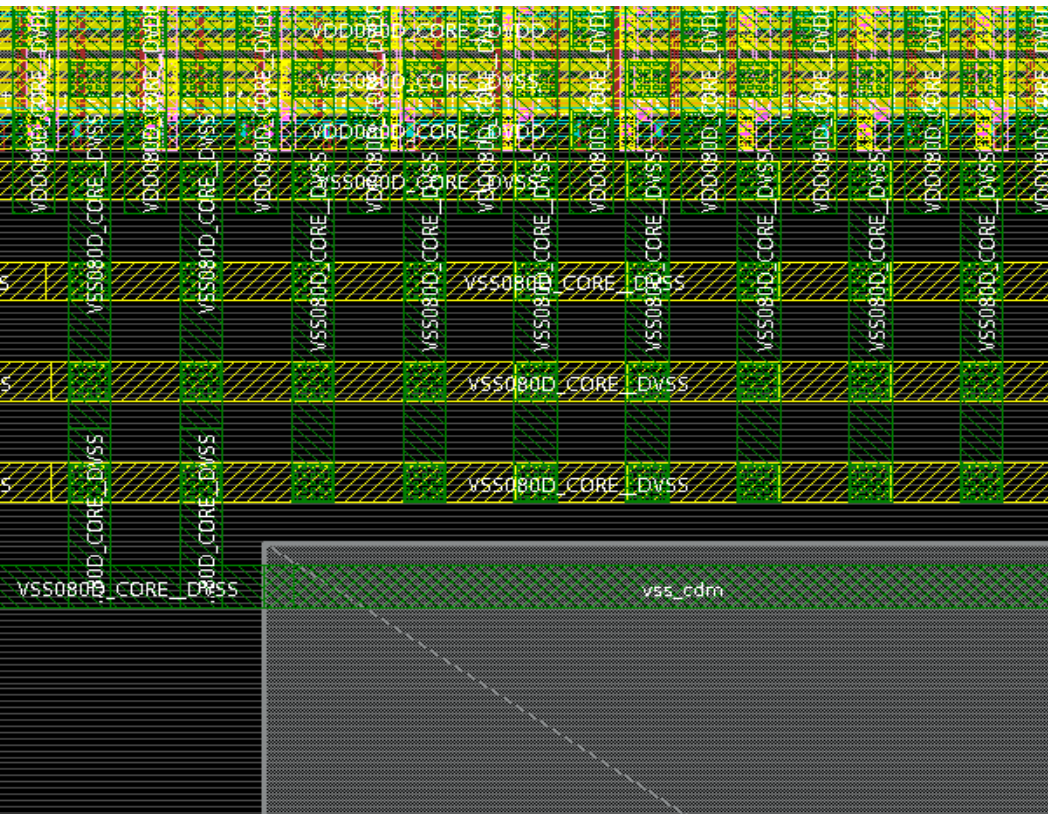


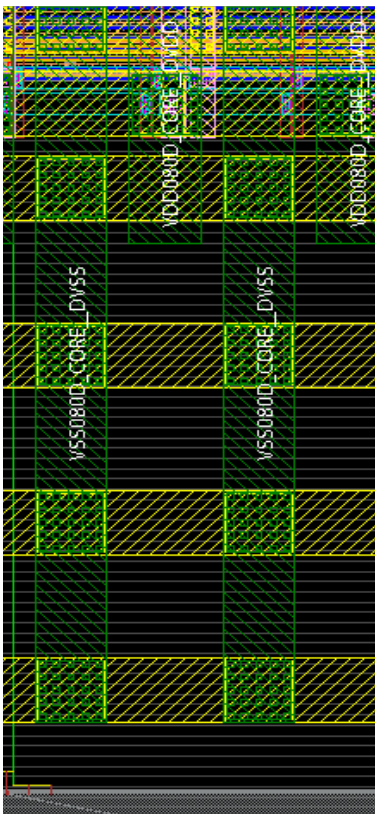


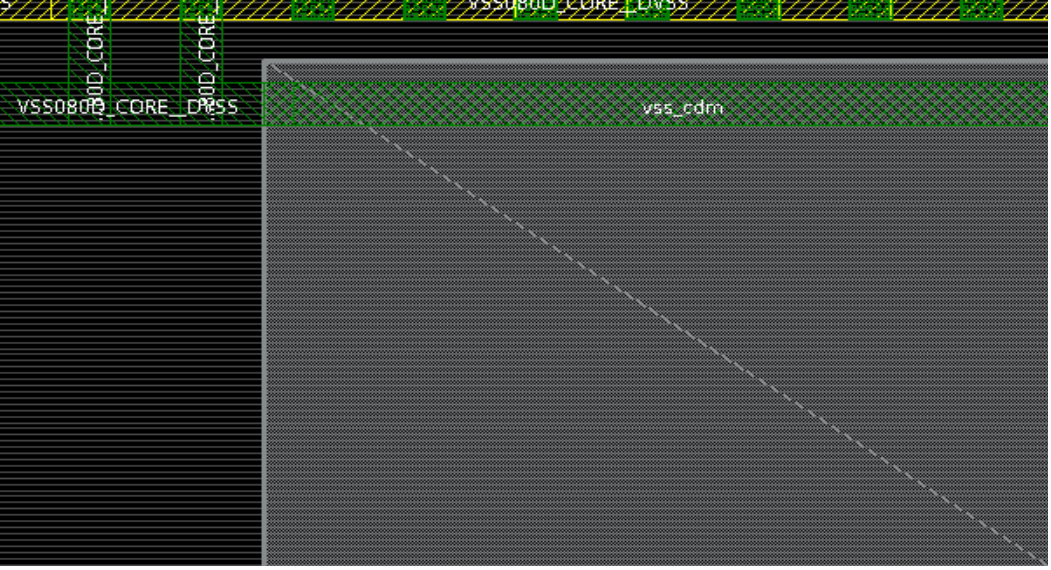
Remark	ETC

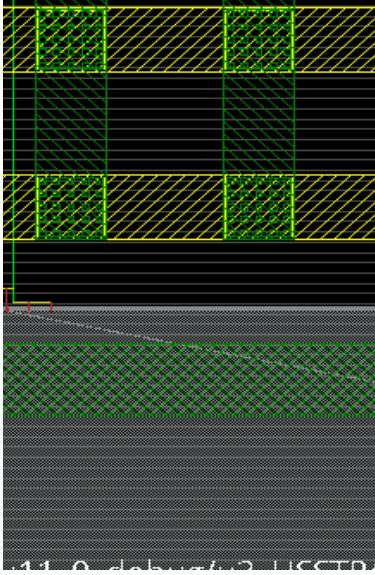
-PERC Result

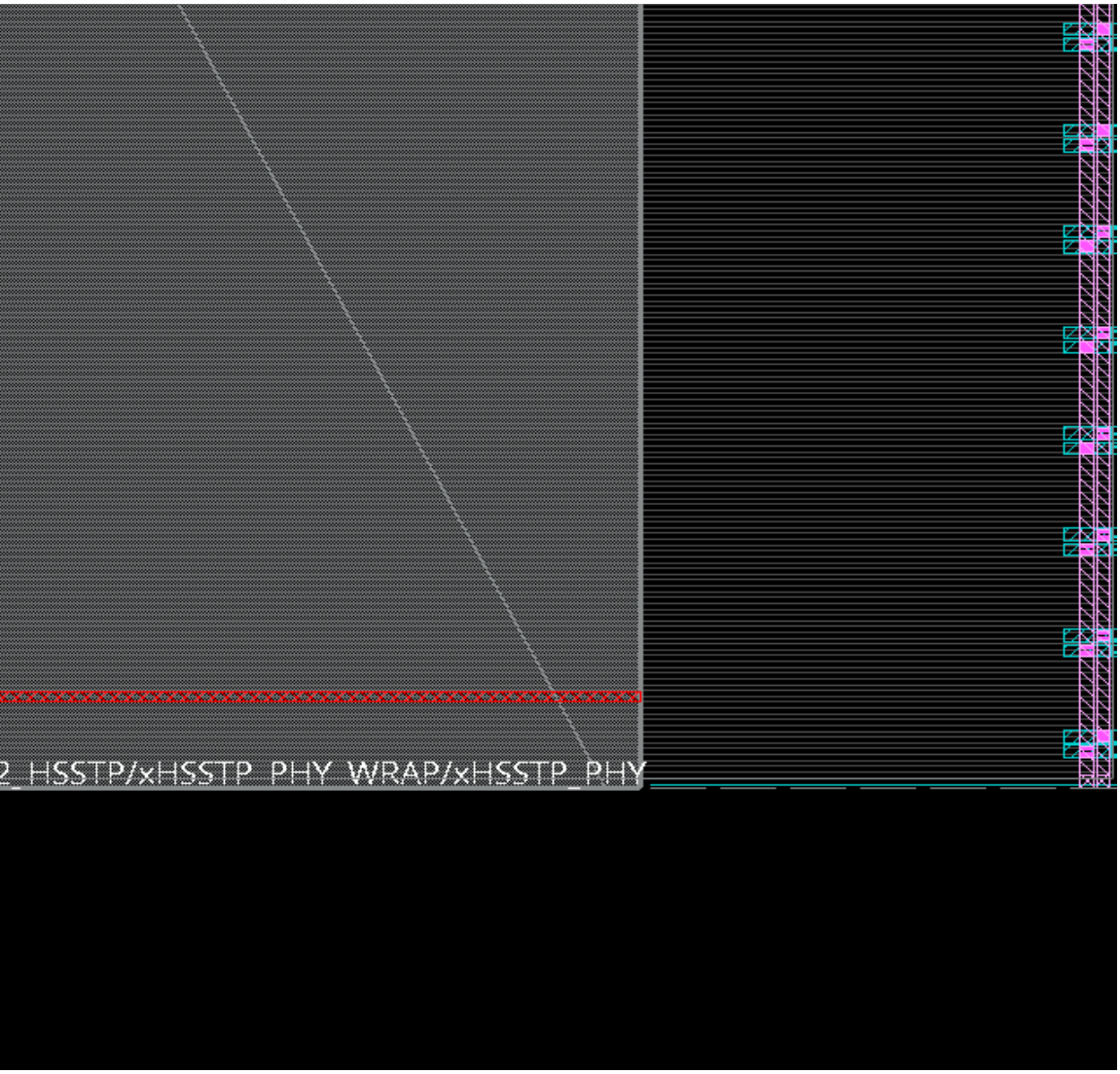
NO	RULE
2	ESDEC41a



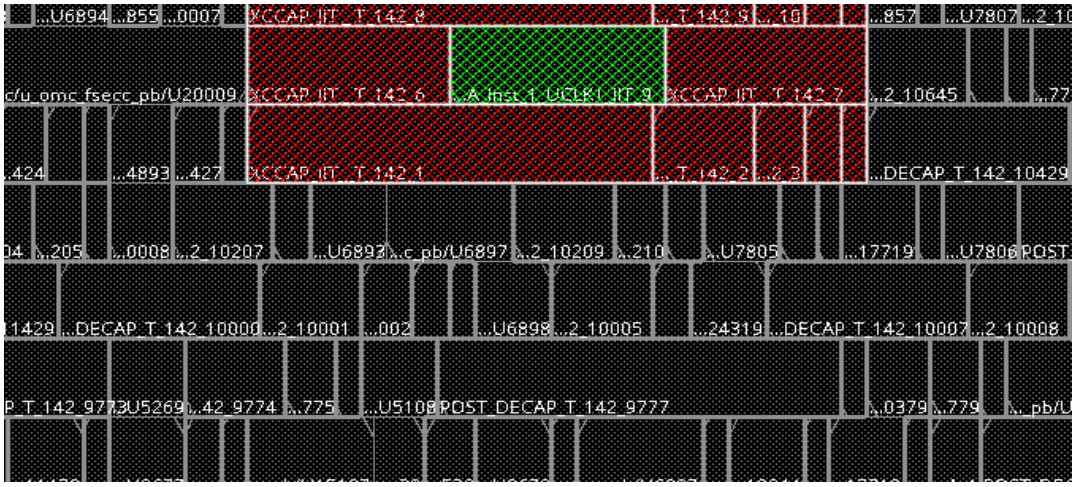



VSS080D_CORE_DVSS
VSS080D_CORE_DVSS
vss_cdm

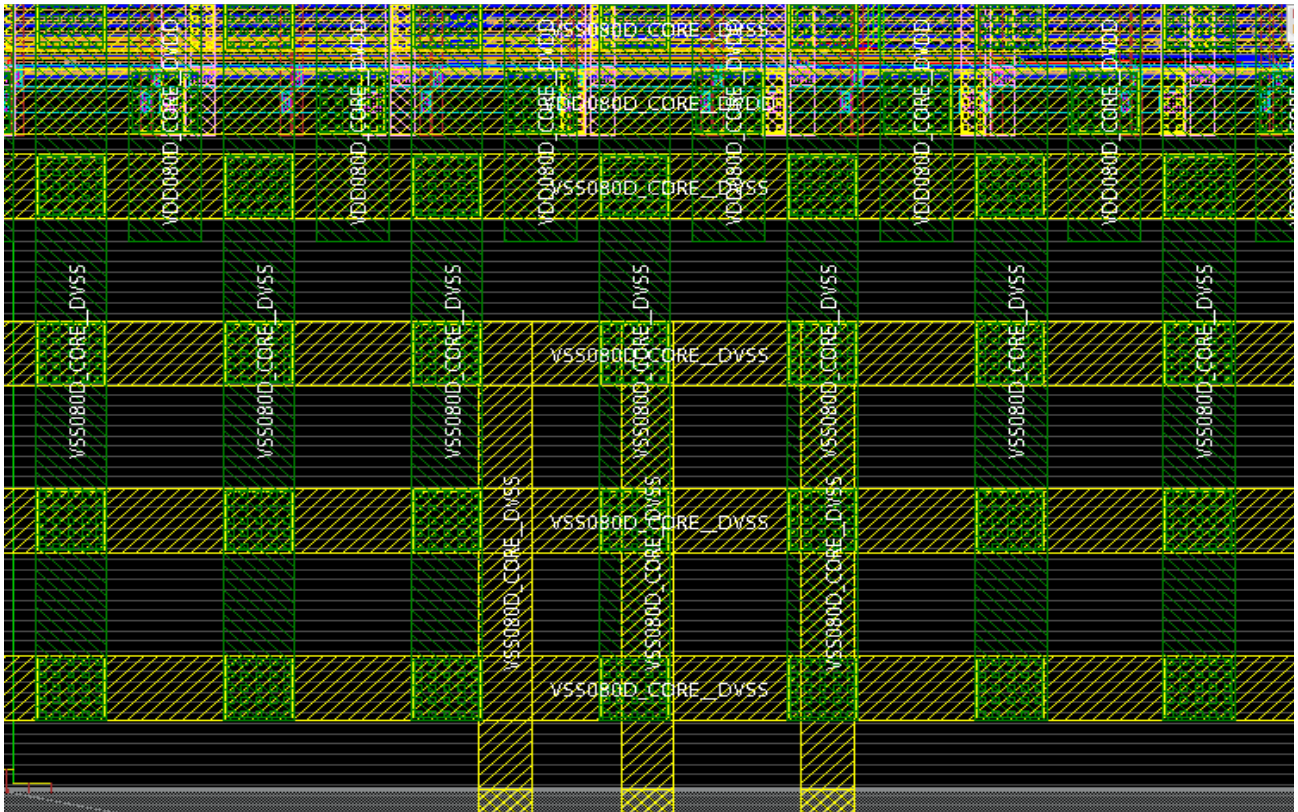


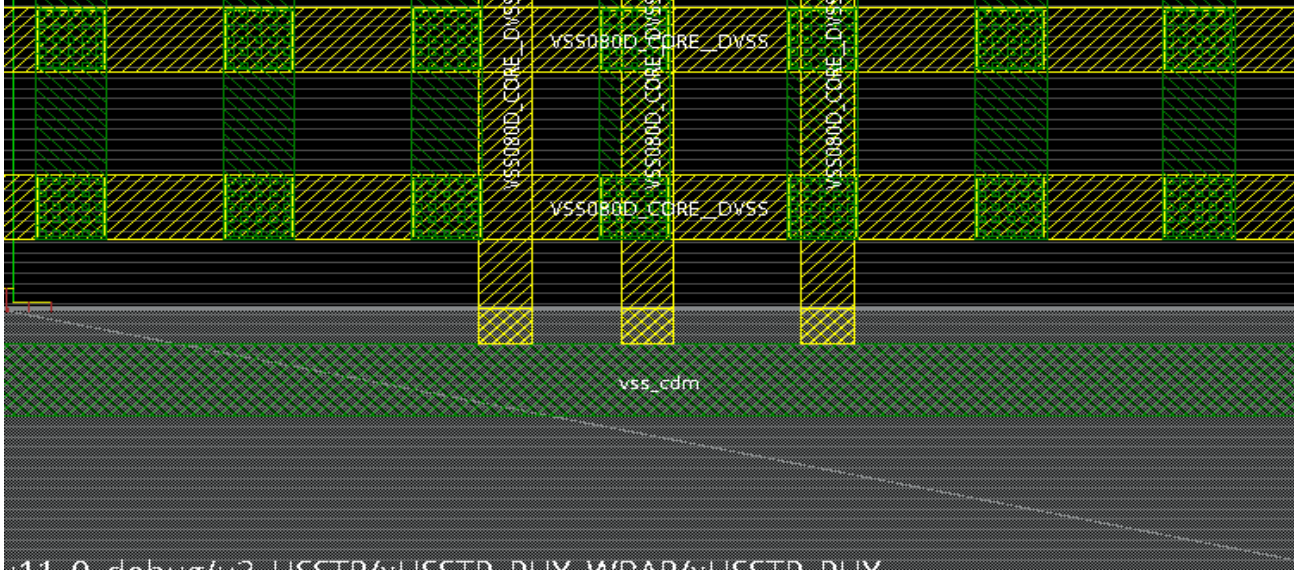

2_HSSTP/xHSSTP_PHY_WRAP/xHSSTP_PHY

- INV With Decap

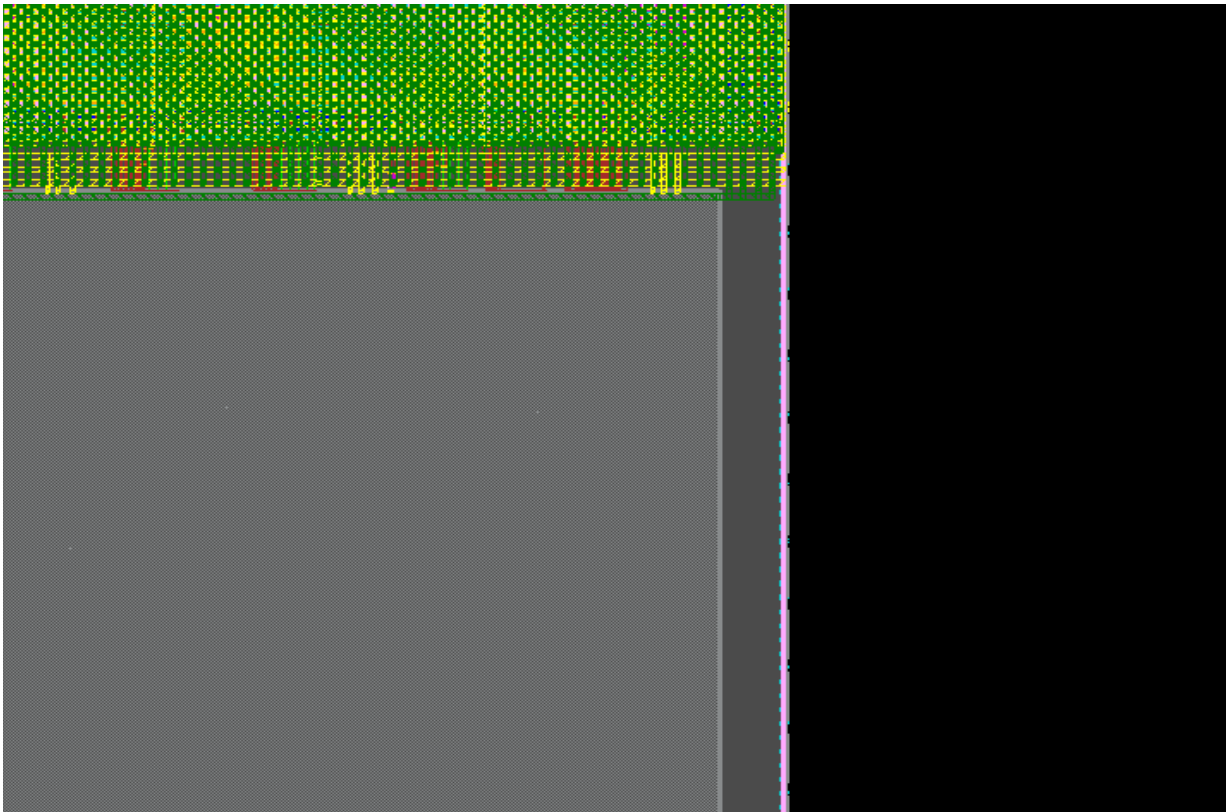

..U6894
..855
..0007
..T_142_9
..857
..U7807
c/u_omc_fsecc_pb/U20009
..2_10645
..424
..4893
..427
..T_142_2
..2_3
..DECAP_T_142_10429
..205
..0008
..2_10207
..U6897
..2_10209
..210
..U7805
..17719
..U7806
..1429
..DECAP_T_142_10000
..2_10001
..002
..U6898
..2_10005
..24319
..DECAP_T_142_10007
..2_10008
..U5269
..42_9774
..775
..U5108
..0379
..779

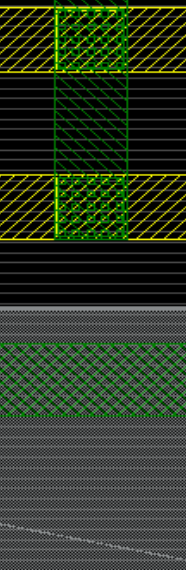
	Description
	0.8V MOSFET between power (gate) / ground (source) / drain




VSS080D_CORE_DWSS
VSS080D_CORE_DWSS
VSS080D_CORE_DWSS
VSS080D_CORE_DWSS
VSS080D_CORE_DWSS
vss_cdm

Number of Error	Comments
1	

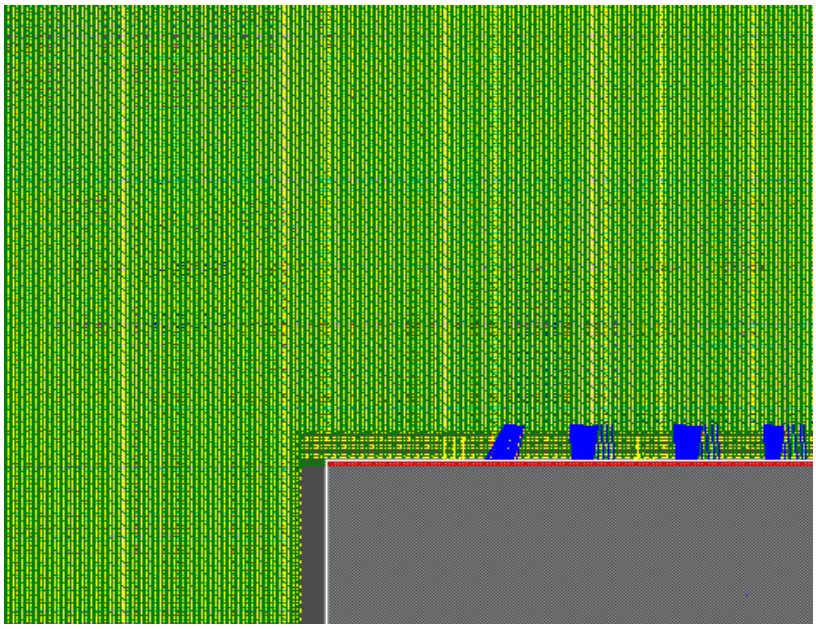


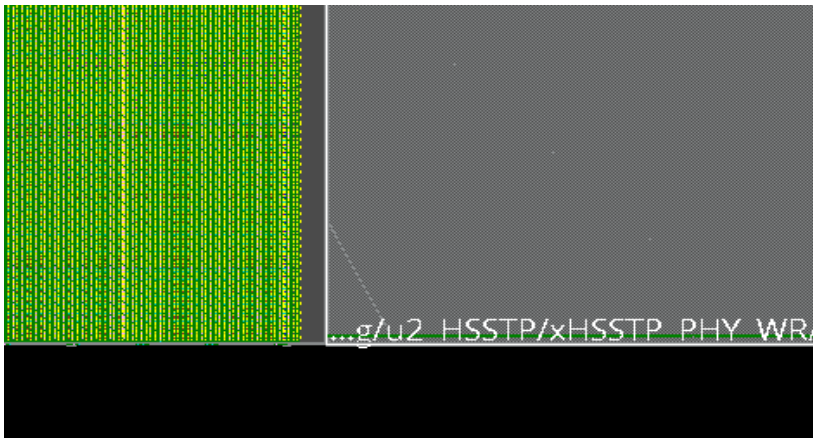


..g/u2_HSSTP/xHSSTP_PHY_WRAP/xHSSTP_PHY

Location	Remark	ETC
rs_hsstptx2ch_ln14lpp_v1p0		

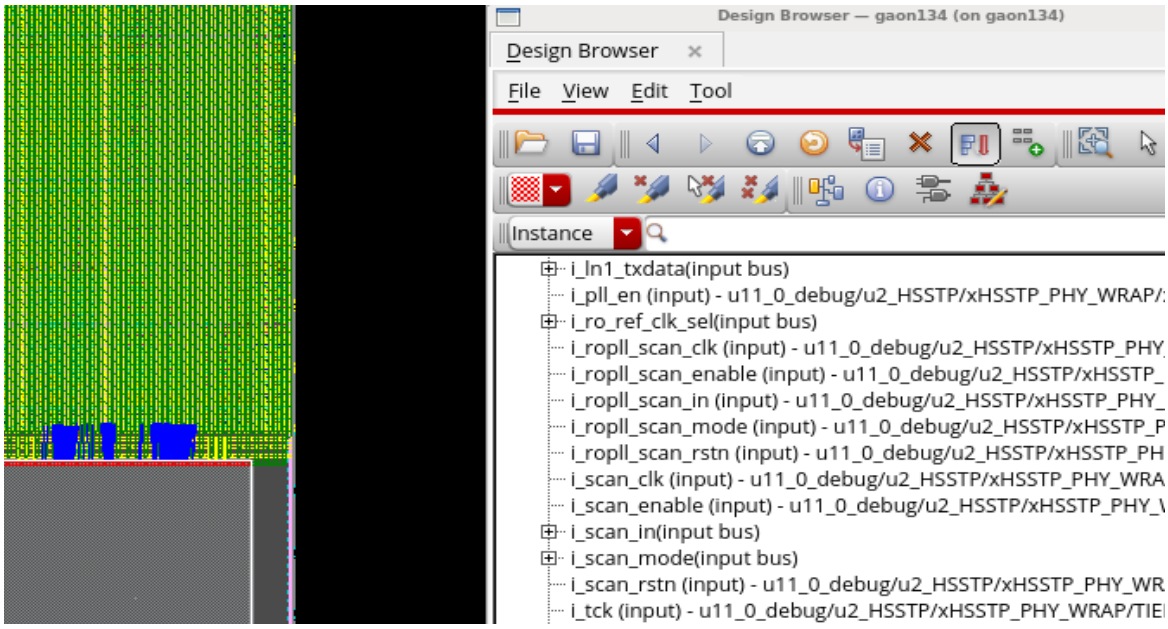
- Power Connection

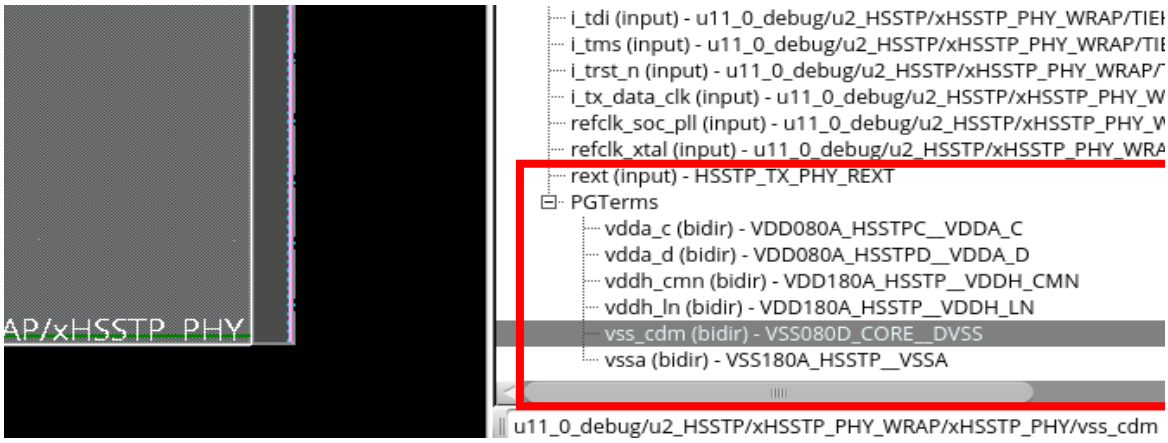


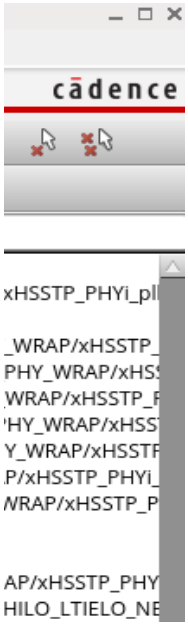


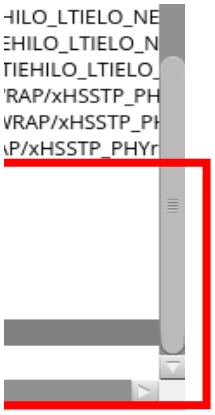
Core POWER Name
VDD080D_CORE__DVDD

Core GROUND Name
VSS080D_CORE__DVSS


Design Browser — gaon134 (on gaon134)
Design Browser
File View Edit Tool
Instance
i_ln1_txdata(input bus)
i_pll_en (input) - u11_0_debug/u2_HSSTP/xHSSTP_PHY_WRAP/
i_ro_ref_clk_sel(input bus)
i_ropll_scan_clk (input) - u11_0_debug/u2_HSSTP/xHSSTP_PHY
i_ropll_scan_enable (input) - u11_0_debug/u2_HSSTP/xHSSTP_
i_ropll_scan_in (input) - u11_0_debug/u2_HSSTP/xHSSTP_PHY_
i_ropll_scan_mode (input) - u11_0_debug/u2_HSSTP/xHSSTP_P
i_ropll_scan_rstn (input) - u11_0_debug/u2_HSSTP/xHSSTP_PH
i_scan_clk (input) - u11_0_debug/u2_HSSTP/xHSSTP_PHY_WRA
i_scan_enable (input) - u11_0_debug/u2_HSSTP/xHSSTP_PHY_
i_scan_in(input bus)
i_scan_mode(input bus)
i_scan_rstn (input) - u11_0_debug/u2_HSSTP/xHSSTP_PHY_WR
i_tck (input) - u11_0_debug/u2_HSSTP/xHSSTP_PHY_WRAP/TIE


AP/xHSSTP_PHY
i_tdi (input) - u11_0_debug/u2_HSSTP/xHSSTP_PHY_WRAP/TIE
i_tms (input) - u11_0_debug/u2_HSSTP/xHSSTP_PHY_WRAP/TI
i_trst_n (input) - u11_0_debug/u2_HSSTP/xHSSTP_PHY_WRAP/
i_tx_data_clk (input) - u11_0_debug/u2_HSSTP/xHSSTP_PHY_W
refclk_soc_pll (input) - u11_0_debug/u2_HSSTP/xHSSTP_PHY_W
refclk_xtal (input) - u11_0_debug/u2_HSSTP/xHSSTP_PHY_WRA
rext (input) - HSSTP_TX_PHY_REXT
PGTerms
vdda_c (bidir) - VDD080A_HSSTPC__VDDA_C
vdda_d (bidir) - VDD080A_HSSTPD__VDDA_D
vddh_cmn (bidir) - VDD180A_HSSTP__VDDH_CMN
vddh_ln (bidir) - VDD180A_HSSTP__VDDH_LN
vss_cdm (bidir) - VSS080D_CORE__DVSS
vssa (bidir) - VSS180A_HSSTP__VSSA
u11_0_debug/u2_HSSTP/xHSSTP_PHY_WRAP/xHSSTP_PHY/vss_cdm


cādence
xHSSTP_PHYi_pll
WRAP/xHSSTP
PHY_WRAP/xHS
WRAP/xHSSTP_
HY_WRAP/xHSS
Y_WRAP/xHSSTP
P/xHSSTP_PHYi_
WRAP/xHSSTP_P
AP/xHSSTP_PHY
HILO_LTIELO_NE


HILO_LTIELO_NE
EHILO_LTIELO_N
TIEHILO_LTIELO_
RAP/xHSSTP_PH
VRAP/xHSSTP_PH
AP/xHSSTP_PHYr